

Governance Framework for Linking Individual-level Health Data

Strategic and Operational Playbook

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¹ OS-PCORTF Project Information: <https://aspe.hhs.gov/creating-new-data-research-environment>.

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1. Overview

Linking individual-level health data across data sources to connect complementary data types increases the patterns and interrelationships that a researcher may explore to accelerate discoveries for prevention, diagnostics, and treatments. Current biomedical research infrastructure lacks a common governance approach that supports linking data for research, including studies of cancer and patient-centered outcomes research. At its highest level, **governance**² is a framework for decision making and clarifying roles and responsibilities based on established policies and processes. Thus, **data governance** refers to the policies, processes, and resulting decisions that impact **data access, sharing, linkage, use, and dissemination**. Without a common approach, researchers that seek to link datasets may find themselves entangled in an expensive and time-consuming exercise of attempting to align the governance processes and requirements of disparate datasets and organizations.

To address these governance challenges, multiple HHS agencies³ collaborated to develop a governance framework that will enable safe and secure linkage of **individual-level health data** for research. The framework is intended for linkages of many data types that entail a high level of governance complexity (hereafter **complex linkages**).⁴ Complex linkages may take many forms, such as requiring an investigator to coordinate multiple asynchronous access requests, navigate several overlapping review processes, or design a linkage implementation plan that aligns with conflicting technical requirements. One complex example would be linking **de-identified** genomic data held in a government's controlled data access repository with electronic health record (EHR) and claims data, each available only through separate secure enclaves.

1.1 Introduction to the Common Governance Framework to Link Individual-level Health Data for Research

This document introduces the Governance Framework to Link Individual-level Health Data for Research (hereafter the **Linkage Governance Framework** or the framework), a structured approach for seeking approval from **dataset holders** and harmonizing governance rules across datasets to enable safe and secure linkage of individual-level health data for research. The Linkage Governance Framework is designed for complex linkages where governance makes the linkage of multiple datasets especially difficult. It aims to accelerate access to data by reducing the burden of adjudicating governance conflicts and gaining required approvals. In order to enable linkage with greater flexibility⁵ and scalability⁶ than is currently available, the framework introduces a new Common Governance Process that:

- Uses a coordinating body as the interface between the investigator, requesting institution, and dataset holders

² Terms found in the glossary are bolded at first significant occurrence.

³ This work was supported through a OS-PCORTF award to the National Institutes of Health (NIH) National Cancer Institute (NCI) and supported the Health FFRDC, operated by MITRE.

⁴ Using PPRL or other technical solutions to discover overlapping study participants from separate data types/datasets in disparate systems to build specific cohorts is out of the scope.

⁵ Flexibility means that the framework is built to facilitate the harmonization of many data types and data collected for purposes other than research.

⁶ Scalability means that the framework is built to create complex linkages for research, public health, and population health applications drawing data from federal, academic, and other institutions.

- Consolidates multiple request forms and entry points into a single request package and centralized entry point
- Synchronizes review processes across dataset holders to occur simultaneously and collaboratively
- Harmonizes governance rules across datasets toward a single rule set to govern the linkage implementation and resulting linked dataset

The Linkage Governance Framework provides a vision and guiding principles to streamline data linkage based on cooperation and collaboration of dataset holders. The Common Governance Process may be applied to any dataset, any data type, and any proposed linkage method and is careful to preserve each dataset holders' critical decision-making authority. By helping to overcome the governance inefficiency and conflicts that can delay research, this framework has the potential to increase the ability to link individual-level health data and accelerate discovery using **linked datasets**.

The Linkage Governance Framework is a proof of concept that requires testing for feasibility, practicality, scalability, and effectiveness. Lessons learned from testing this new approach and process will be used to further refine and enhance the framework with the eventual goal of full-scale implementation and widespread availability across research communities. If implemented, potentially along with other technological innovations,⁷ the Linkage Governance Framework could represent a significant innovation for data linkage and research by increasing data reuse and accelerating scientific discovery.

1.2 Purpose of the Playbook

This playbook is a structured guide designed to support understanding, adoption, and implementation of the Linkage Governance Framework. It outlines the framework's purpose, objectives, guiding principles, and the Common Governance Process. Within the process, the playbook defines the roles and responsibilities, governance decisions, and implementation tools to support the framework's use. Together, the contents of this playbook support Linkage Governance Framework adoption by:

- Fostering a shared understanding of the Common Governance Process and expectations of participants, enabling individuals to understand how the framework may be used in real-world scenarios;
- Ensuring that all elements of the framework adhere to a unified vision, reducing ambiguity and fostering coherence;
- Promoting a transparent governance process and decision making for data linkage in research; and
- Illustrating how a process can facilitate efficient governance decision making aligned with the framework's guiding principles.

As the framework matures beyond a proof of concept, the playbook will evolve and expand into an implementation guide and the technology to support the framework and Common Governance Process will be developed. In the future, the framework will likely be anchored by a central technology platform that holds framework documentation, supports process functions,

⁷ For further information on complementary efforts, access the OS-PCORTF Project Information: <https://aspe.hhs.gov/creating-new-data-research-environment>.

catalogs governance decisions, and provides an interface for investigators, Requesting Institutions, and dataset holders.

1.3 Structure of the Playbook

The playbook can be read in its entirety or referenced to understand specific topics. The structure of the playbook reflects the framework's design, placing the Common Governance Process at its center and presenting the various facets – such as roles and responsibilities – as integral elements. As such, the document is organized so that a reader can easily identify the relevant section that addresses a particular subject or issue:

- Background ([Section 2](#)): Level-sets on key concepts – the meaning of linking individual-level health data, why it is important, and why a Linkage Governance Framework is needed.
- Framework Overview ([Section 3](#)): Establishes the framework's purpose, objectives, elements, and guiding principles.
- Common Governance Process ([Section 4](#)): Describes and maps the sequence of governance activities.
- Roles and Responsibilities ([Section 5](#)): Outlines the essential positions and functions for the framework.
- Governance Decisions ([Section 6](#)): Discusses decisions and their inputs and outputs.
- Implementation Tools ([Section 7](#)): Describes tools that support framework adoption.
- Continuous Improvement ([Section 8](#)): Describes the three activities essential for capturing and applying lessons learned.

1.4 Audience

The playbook is a flexible resource intended for multiple audiences. The primary audience for this playbook is **investigators**, **requesting institutions**, and dataset holders who will participate in the Common Governance Process as well as the **coordinating body** that will operate the Linkage Governance Framework. Secondary audiences may include research funders, oversight entities, industry, and other members of the research ecosystem who may use this playbook to explore an alternative governance approach.

1.5 How to Use the Playbook

Readers can consult the playbook for a comprehensive understanding of the Linkage Governance Framework and its implementation through the Common Governance Process. Readers should refer to the playbook to understand how the Common Governance Process is designed to operate and the governance decisions it aims to facilitate. They may also consult the playbook to discover the available implementation tools.

2. Background

2.1 What Does Linking Individual-level Health Data Mean?

Linkage and linking data are not terms with single consensus definitions. This framework is focused on linking individual-level health data, which is defined as health-related person-level data that is found in electronic health records (EHRs), research repositories, registries, and other data systems and may or may not include **protected health information (PHI)** or **personally identifiable information (PII)**.

An individual (patient) often has relevant health data in multiple independent datasets. These **“originating” datasets** can be EHRs that hold clinical records from doctor or hospital visits, national or state health tracking systems such as cancer registries, or research data repositories that hold data from participation in research studies. Individual-level linkage in this framework refers to bringing the information present in disparate originating datasets together at an individual level within a resulting linked dataset. This requires recognizing or matching the same individual across the originating datasets. Many methods exist to match individuals across the originating datasets. Some methods require shared identifiers such as names, addresses, dates of birth, and social security numbers. Other methods use common de-identified tokens that are generated from identifiers and known only to the dataset holder and an honest broker (e.g., privacy-preserving record linkage or PPRL). For either method, the resulting linked dataset may or may not remove all PII and/or PHI from the originating datasets.

By combining data elements at the individual level from all originating datasets, the resulting linked dataset will contain more complete information about an individual. However, the resulting linked dataset will not necessarily include all data elements present in originating datasets. Additionally, the resulting linked dataset will usually include fewer individuals than those present across the originating datasets. Each originating dataset has an independent set of participants so complete information in the resulting dataset will only be available for those individuals present in all originating datasets (the overlap of the originating datasets). The number of individuals in the resulting linked dataset may be further reduced when applying inclusion and/or exclusion criteria such as a specific type of cancer or a desired age range. The size of the study population (number of individuals in the linked dataset) is of crucial importance to the investigator’s ability to draw statistically meaningful conclusions.

Once created, linked datasets may be linked with additional data (subject to the governance rules of the linked datasets) to create subsequent linked datasets. As such, data that investigators want to link may already be a product of previous linkages. These further linkages are only possible if identifying information for the individual is maintained either in the linked dataset or by an honest broker for the linked dataset.

2.2 Why Is Linking Individual-level Health Data for Research Important?

Linking individual-level health data drives innovation and discovery in research by expanding both access to and the utility of health data. This integration enhances analytical power and opens the possibility of new research insights by revealing patterns and relationships that may not be apparent in a single dataset. These insights have the potential to accelerate breakthroughs in disease causes, prevention strategies, diagnostic approaches, treatments, and cures. Key examples of the value of individual-level health data linkage include:

- **More complete understanding of factors contributing to disease:** Data linkage combines data from diverse sources to create a more comprehensive view of health at an individual level and enable deeper analyses of risk factors, disease states, and outcomes.
- **Longitudinal analysis:** Data linkage provides a mechanism to track health outcomes over time outside traditional, consented research studies, potentially yielding insights into disease progression, treatment effectiveness, long-term impacts of interventions, and trends in chronic conditions and population health.
- **Additional population health insights:** Data linkage can reveal patterns across demographic, geographic, and genetic risk factors and expand the size of study populations to enable detection of subtle inferences or rare variations.
- **Catalyst for innovative research:** Data linkage facilitates creation of comprehensive datasets that support complex analyses and foster discovery through innovative methodologies such as predictive modeling and machine learning.
- **Less redundant research:** Data linkage reduces costly redundant **data collection** and data generation in research such as repeated genetic sequencing.

2.3 Why Is a Linkage Governance Framework Needed for Research Studies Linking Individual-level Health Data?

While linking individual-level health data unlocks the potential to advance scientific discoveries, it also introduces governance challenges that must be addressed before research can begin. Each originating dataset or data source is governed by its own set of policies that specify rules⁸ for how investigators can access, share, link, use, and disseminate data within or between organizations (referred to as the **data lifecycle**). Unlike a study where an investigator accesses and analyzes a single dataset, a study including data linkage means an investigator will require access to multiple originating datasets and from them will create a new, distinct linked dataset to analyze. When datasets are linked, the **governance rules** from each originating dataset are inherited by the resulting linked dataset. In order for linkage to be permitted, governance rules from disparate sources must be harmonized to resolve conflicts.

Governance complexity related to linking datasets can delay research and result in inefficiencies that increase costs, especially in investigator effort and administrative time. Navigating multiple governance processes for each dataset is inherently time-consuming for investigators, with each process having distinct requirements, application forms, approval procedures, and timelines. Investigators must obtain approvals and fulfill policy and process requirements from each dataset holder independently. When a dataset holder reviews an application for data access and linkage, they not only evaluate how investigators will use their data but also consider how their data will interact with other datasets. This not only duplicates effort, but it creates uncertainty about whether linking data creates excess risk and whether established privacy and security controls will remain effective in the context of the linked dataset.

Even when data linkage appears feasible and permissible, and aligns with privacy and security requirements, the process can stall without mechanisms to reconcile or harmonize conflicting

⁸ Governance rules are established by governance policies and originate from a variety of sources, including laws; institutional policies; project-, study-, or dataset-specific policies; ethical requirements; and organizational practices and norms. Each source can hold one or multiple policies that assert a range of rules from broad principles for overall data management to highly specific requirements targeting a single aspect of the data lifecycle.

governance rules. Without a clear framework that creates governance rules that are mutually acceptable to all dataset holders, investigators may face contradictory requirements that multiply as the number of datasets to link grows. This may derail research, leading investigators pursuing a complex linkage to abandon the research.

Streamlining these processes through a framework could reduce the time and costs associated with requesting data access, obtaining approvals, and executing **data use agreements** or **DUAs**, allowing investigators to direct resources instead to research and discovery. Shorter timeframes to complete required governance work and clearer information on how to obtain required approvals also lower operational costs for dataset holders, allowing them to allocate resources more effectively.

3. Framework Overview

3.1 Purpose

The Linkage Governance Framework establishes an innovative, replicable, and transparent approach to the governance for linking individual-level health data. This framework was created to increase efficiency in data linkage for research by reducing administrative burden, thereby accelerating research timelines and access to data. Added efficiency will make it easier, faster, and more cost-effective for investigators, requesting institutions, and dataset holders to complete the governance activities that facilitate access to data and successful linkage implementation, and launch research. This proof-of-concept framework is intended for complex linkages of health data for research purposes. It is designed to be scalable⁹ and flexible,¹⁰ enabling the integration of data of all kinds that originate from federal, state, local, and non-government sources.

To achieve this purpose, the framework introduces a Common Governance Process that enables safe and secure linkages by helping investigators, requesting institutions, and dataset holders make governance decisions quickly and navigate overlapping or conflicting governance rules efficiently and collaboratively. The process includes three stages dedicated to gathering linkage information, screening linkage requests for feasibility, and harmonizing governance rules across datasets while honoring the original intent behind each governance policy. While this process endeavors to accelerate governance agreements (e.g., DUAs), it defers to the organizations involved in each study to determine the best approach to setting up and executing those agreements.

3.2 Objectives

Clearly defined objectives help communicate the value of the Linkage Governance Framework. The near-, mid-, and long-term objectives below describe the results-oriented objectives that a Linkage Governance Framework may be able to achieve.¹¹

⁹ Scalable governance is built to create data linkages for research, public health, and population health applications drawing data from federal, academic, and other institutions.

¹⁰ Flexible governance is built to facilitate the harmonization of many data types and data collected for research and other purposes.

¹¹ As this is a proof-of-concept framework, the objectives do not currently include measurable and time-bound parameters. As the framework evolves, measurable parameters will be important for tracking performance and progress.

Near-term objectives

- More research studies linking individual-level health data across HHS datasets are initiated.
- Complex linkages are executed in shorter timeframes and more efficiently due to consolidating governance activities, reducing steps, and minimizing delays.
- Fewer complex linkage requests are abandoned due to governance obstacles.

Mid-term objectives

- More investigators complete research studies using linked data due to consensus building and clarity across Datasets Holders and investigators.
- Research costs are reduced by streamlining governance and enabling reusable pathways for health data linkage.

Long-term objective

- More health discoveries happen faster.

3.3 Elements

The framework includes five elements (see Figure 1) that support a structured approach to overcoming governance barriers and facilitating complex linkages of individual-level health data for research:

1. **Guiding Principles:** Articulates the core values that underpin the framework's purpose and objectives.
2. **Common Governance Process:** Provides consistent, transparent processes to intake, screen, and harmonize a request to access and link multiple datasets.
3. **Roles and Responsibilities:** Defines the functions of each role involved in the governance process and outlines specific duties associated with each.
4. **Governance Decisions:** Specifies the key decisions that must be made to achieve the framework's objectives.
5. **Continuous Improvement:** Builds governance effectiveness through structured feedback to refine and enhance the framework over time.



Figure 1: Framework Elements

3.4 Guiding Principles

The Linkage Governance Framework endeavors to enable safe and secure data linkages that have been difficult to create in the past. These complex linkages are made possible by providing a forum and processes for dataset holders to collaborate and by fostering trust, transparency, and mutual recognition around the potential value of linking individual-level health data for research and scientific discovery. These guiding principles articulate the core values that drive

the framework's purpose, processes, and objectives:

Principle 1. Ensuring access: Investigators must have access to linked individual-level health data, as it is essential for advancing research that improves the lives of all Americans.

Principle 2. Balancing linkage with protection of data: Dataset holders must balance advancing data linkages for research while safeguarding individual privacy, ensuring data security, and complying with applicable laws and policies.

Principle 3. Encouraging a patient-centered approach: Governance decision making must incorporate patient perspectives whenever possible, and dataset holders' patient representatives are important participants in the Common Governance Process.

Principle 4. Upholding authority and accountability: The Linkage Governance Framework upholds the authority(-ies) of dataset holders, investigators, and requesting institutions:

4a. Dataset holders are responsible for decisions about access, sharing, linkage, use, and dissemination of their data.

4b. Investigators and requesting institutions are responsible for conducting research and managing associated risk.

Principle 5. Advancing transparency and collaboration: Effective governance relies on decision making that is transparent, collaborative, and cooperative to facilitate data linkage for research.

Principle 6. Ensuring documentation and reproducibility: The Linkage Governance Framework requires the documentation of inputs, decisions, and outcomes to support the reproducibility and integrity of research findings and future efforts.

Principle 7. Promoting continuous improvement: The Linkage Governance Framework evolves through continuous improvement to adapt to changing needs and enhance governance effectiveness.

4. Common Governance Process

Within the Linkage Governance Framework, the Common Governance Process operationalizes the framework's guiding principles ([Section 3.4](#)) within a sequence of governance activities based on defined roles and responsibilities ([Section 5](#)) and governance decisions ([Section 6](#)) with support from a set of implementation tools ([Section 7](#)). In keeping with the core values expressed in the guiding principles, the process aims to streamline governance information gathering and analysis, align dataset holders review processes, and create a forum for dataset holders to collaborate as they attempt to harmonize governance rules and either approve or deny a request from an investigator and requesting institution. A set of standards supporting the common governance processes are described in [Section 7.1](#).

Importantly, the Common Governance Process sits within the larger research lifecycle. For a given study, the research lifecycle starts with the investigator's inspiration for and design of a

research study and proceeds through a series of governance activities (e.g., reviews, approvals, and agreements); the approved study launches, runs its course, and ends with study closeout and disposition of the study materials.

The Common Governance Process transects this lifecycle as a part of the governance activities and is organized into three stages: Intake, Screening, and Rule Harmonization (Figure 2). Each of the three stages has a dedicated section below that describes its activities, the roles involved, and the governance decisions made in that stage. [Appendix C](#) presents twelve standards defined to translate the framework's guiding principles to an operational level and guide the Common Governance Process. Business process diagrams are included at the end of Section 4 to illustrate who is responsible for specific parts of the work within each stage, and relevant implementation tools are referenced accordingly.

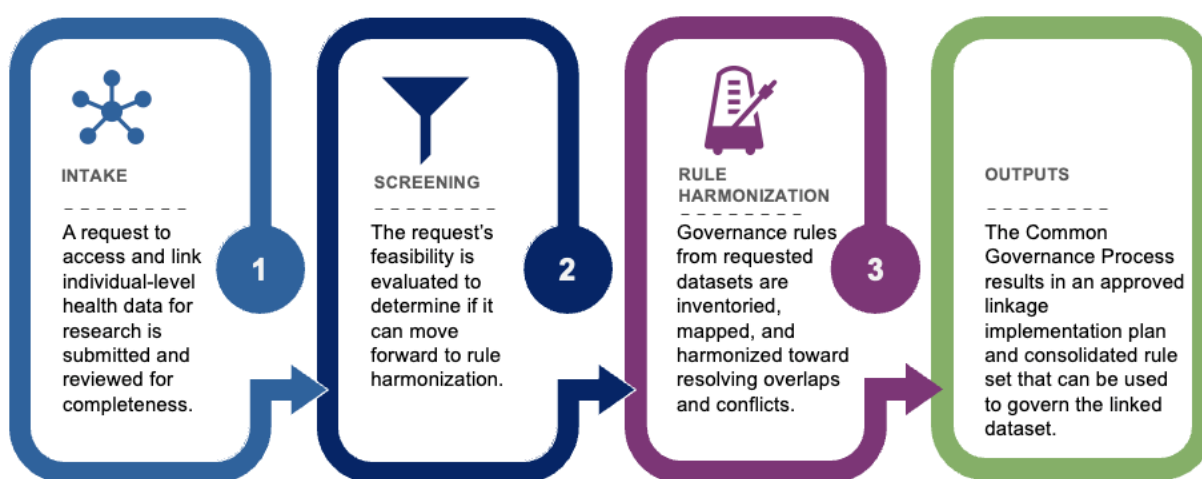


Figure 2: Common Governance Process Overview

4.1 Intake

Intake begins when the investigator/requesting institution¹² submits a request package to the coordinating body (see 1.0-1.2 in Figure 5). This first stage in the Common Governance Process aims to gather all information needed to make governance decisions within Screening and Rule Harmonization. Before beginning a request package, the investigator/requesting institution are expected to have designed the study, defined an approach and methods to link data, completed **institutional review board (IRB)** and **privacy board** reviews as required and received approvals, and completed any other Institution-specific approvals.

The investigator/requesting institution are also expected to have familiarized themselves with the Common Governance Process and reviewed the governance for the datasets they are requesting to the degree they are accessible.

A request package includes six steps: request creation, study and administrative document

¹² Because research operations vary across requesting institutions, the framework combines the responsibilities of the investigator and requesting institution, indicated by "investigator/requesting institution," so that the delegation of responsibilities may reflect the policies and practices of the requesting institution.

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upload, dataset request form, dataset-specific governance questions, attestations, and Request Package affirmation for submission (see Figure 3).¹³ This framework and process anticipate that a central platform will be developed to allow the investigator/requesting institution to create the request record and the Request Package and help the coordinating body manage the request process flow, automate communications, and house information on the outcomes of the Common Governance Process.¹⁴

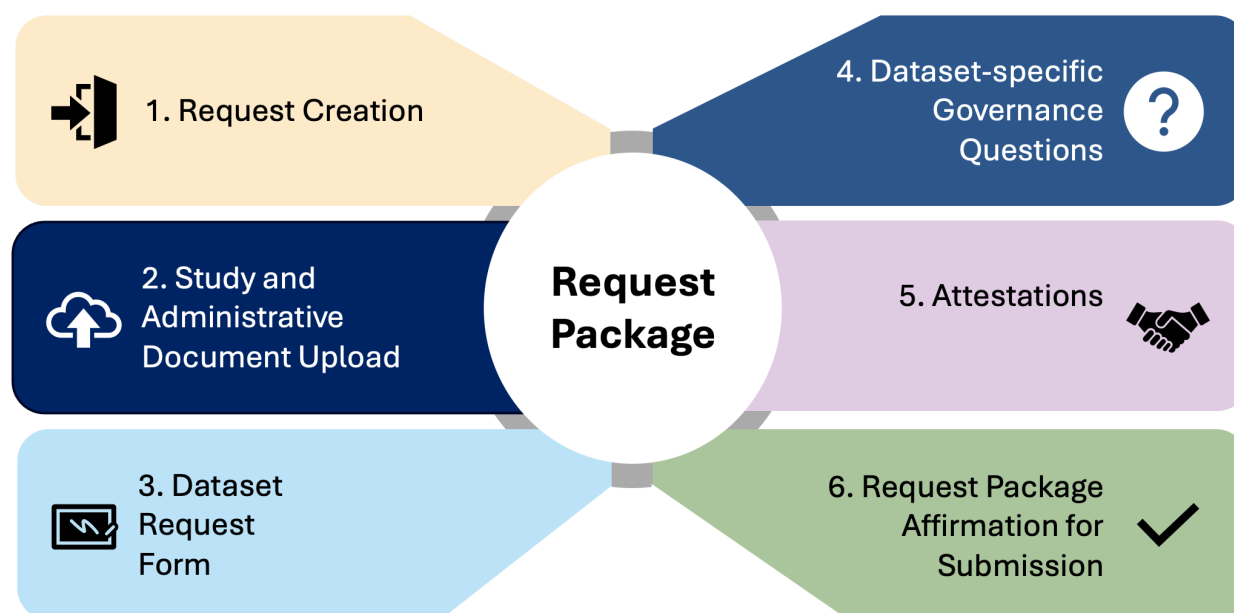


Figure 3: Request Package Steps

To complete the Request Package, the investigator/requesting institution first creates a request, which generates a record that will be used to track request progress and decisions throughout the governance process. The investigator/requesting institution upload study and administrative documents such as study and IRB protocols and approvals from the IRB and privacy board and complete the Dataset Request Form that collects information about:

- Study aims, analysis, duration, and funding;
- Organizations and individuals participating in the study;
- The IRB and Privacy Board of record affiliated with uploaded study and administrative documents;¹⁵
- Requested datasets and **linkage implementation** (i.e., how the linkage will be carried out);
- Creation of, access to, use of, and disposition of the linked dataset; and
- Privacy and security of data throughout the study.

¹³ See [Appendix D](#) for a detailed description of request package contents and discussion of required activities that precede request package submission.

¹⁴ Defining such a platform is beyond the scope of this project.

¹⁵ For applicable studies, an explanation of why IRB and Privacy Board review is not necessary may be provided in lieu of IRB and Privacy Board information and documentation.

Before the request package can be submitted, the investigator/requesting institution must complete attestations that affirm the relationship between the investigator and requesting institution, confirm that they have done the expected pre-work, confer the accountability of request package information to the investigator and requesting institution, and verify the accuracy of information in the request package.

Once submitted, the coordinating body reviews the request package for completeness and makes an operational decision about whether the request package is ready for dataset holder review (see 1.3-1.5 in Figure 5). Once confirmed for completeness, the coordinating body advances the request package to the Screening stage (see 1.11 in Figure 5). If incomplete, the coordinating body returns the request package to the investigator/requesting institution to be revised or not (see 1.6-1.10 in Figure 5).

4.2 Screening

Screening is the second stage in the Common Governance Process and aims to:

1. Transmit the completed request package to the dataset holders;
2. Identify the reviewer(s) from the dataset holders that will evaluate the request package; and
3. Confirm that the request is feasible and can advance to rule harmonization.

Screening includes activities carried out by investigator/requesting institution, coordinating body, and dataset holders (see Screening Business Process Diagram, Figure 6). The governance decision made in Screening sets the stage for rule harmonization, ensuring that only feasible requests move on to rule harmonization.

The Screening stage begins once the request package is confirmed by the coordinating body to be complete. The coordinating body shares the request package with the dataset holders for the requested datasets noted in the request package and requests assignment of a reviewer (see 2.0-2.3 in Figure 6). The dataset holder-assigned reviewer(s) should be the individual(s) who have strong knowledge of the requested dataset and is willing and authorized to make (or coordinate) timely decisions regarding the request package in this stage and in the subsequent Rule Harmonization stage. Reviewers will likely need to engage other groups (e.g., Data Access Committee) or other individuals (e.g., chief medical officer, policy director, patient representative) within their organization to make governance decisions within Screening and Rule Harmonization.

The dataset holder notifies the coordinating body (see 2.4-2.6 in Figure 6) of the reviewer assignment. If a timely notification is not made, the coordinating body is responsible for repeating the request (see 2.7 in Figure 6). The assigned reviewers then screen the request package and make a “go/no-go” decision on whether the proposed study and use of data is feasible enough to move on to rule harmonization (see 2.5-2.10 in Figure 6). While the Process relies on dataset holders to interpret feasibility for the datasets that they represent, suggested feasibility questions might include:

- Are the data attributes present in the requested datasets?
- Is the intended study population present in the requested datasets?
- Is the proposed linkage implementation technically feasible as described?

- Are there any “red flags” that would be impossible to resolve in rule harmonization?

An assessment of feasibility does not include concerns regarding the alignment of governance rules across datasets that can be adjudicated in the subsequent Rule Harmonization stage.

If the request package is feasible, then the dataset holder sends a “go” decision to the coordinating body (see 2.12 in Figure 6). If it is not feasible, the dataset holder sends a “no-go” decision to the coordinating body along with feedback on why the study is not a “go” and what could be changed to make it feasible (see 2.11 in Figure 6). This marks the completion of the first governance decision of the Common Governance Process (see 2.10 in Figure 6), wherein each dataset holder’s decision is based on their dataset’s unique features and policies.

Once go/no-go decisions have been received from all dataset holders, the coordinating body reviews the decisions and feedback from dataset holders in order to determine whether the request may proceed to rule harmonization (see 2.9-2.20 in Figure 6). If any dataset holder returns a “no-go,” the request may not proceed, and the coordinating body notifies the investigator/requesting institution and provides feedback so that they may determine whether the study will be revised and resubmitted (see 2.15-2.18 in Figure 6). If all dataset holders returned a “go,” the coordinating body notifies the investigator/requesting institution that the request is moving forward to Rule Harmonization (see 2.21-2.25 in Figure 6).

4.3 Rule Harmonization

Rule Harmonization is the third and final stage in the Common Governance Process (see Rule Harmonization Overview in Figure 4 and Business Process Diagrams in Figure 7, Figure 8, and Figure 9 and a Rule Harmonization Protocol in [Appendix E](#)). The aims of rule harmonization are to:

1. Assemble and organize a complete inventory of requested datasets’ applicable governance rules;
2. Map and categorize governance rules, including identifying those that require harmonization;
3. Draft a single set of rules that applies to a linked dataset and reflects the governance rules from the originating datasets; such a **consolidated rule set** includes harmonized rules that resolve rule overlaps and conflicts;
4. Resolve any issues of concern with the draft consolidated rule set through collaboration and cooperation; and
5. Issue a final approval or denial of the request.

Rule harmonization includes activities carried out by the investigator/requesting institution, the coordinating body, and the dataset holders. Four governance decisions are made during rule harmonization that together determine whether the request is approved and, if approved, the consolidated rule set that will guide how the linkage implementation will occur and how the resulting linked data may be accessed, shared, used, and disseminated. Rule harmonization begins once the dataset holders have confirmed that the request is feasible.

The coordinating body collects the governance sources from each dataset holder that contain the policies relevant to the requested dataset’s linkage and use for research. The coordinating body reviews each source and extracts the relevant governance rules, organizing rules into a

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draft rule inventory (see 3.0-3.2 in Figure 7). Each rule is classified into a domain and subdomain. Because governance sources often reference other sources (e.g., a DUA references the Health Insurance Portability and Accountability Act [HIPAA]), the coordinating body may identify and retrieve additional sources while creating the rule inventory.

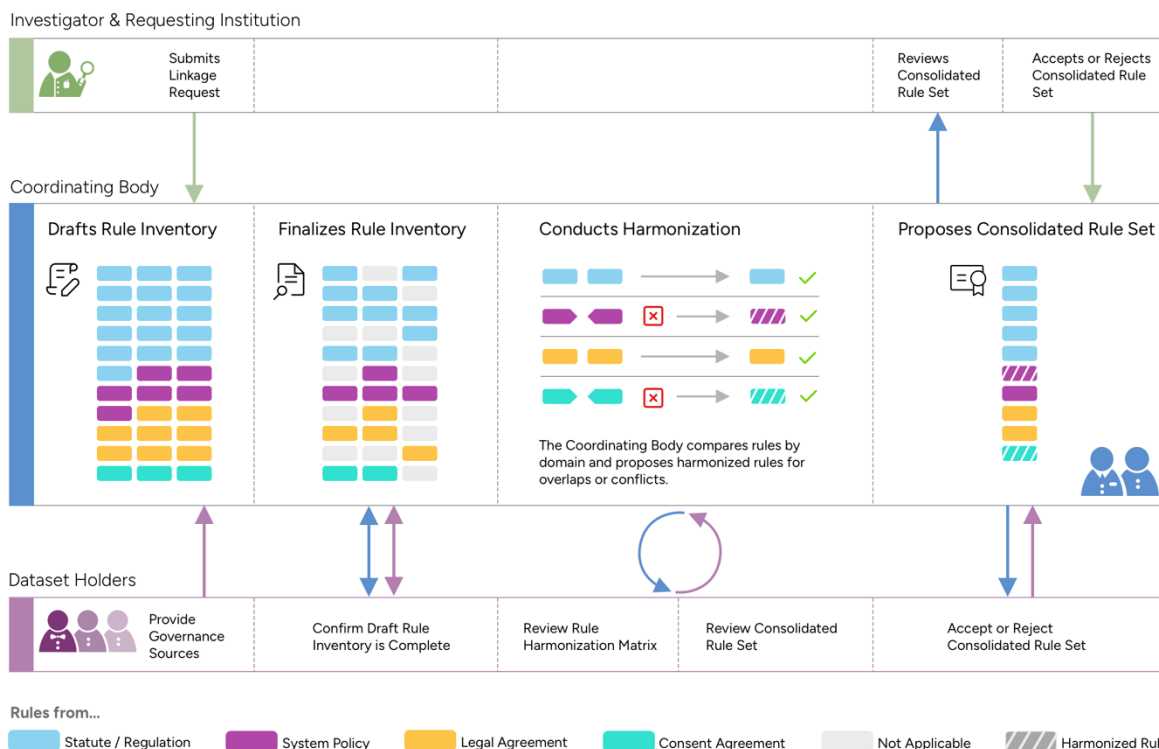


Figure 4: Rule Harmonization Overview

Once complete, the coordinating body sends each dataset holder the draft rule inventory to validate that all applicable rules are present and that rules have been classified accurately (see 3.3-3.6 in Figure 7). The draft rule inventory may go through multiple revisions until it is accepted by dataset holders (see 3.7-3.10 in Figure 7). This marks the second governance decision in the Common Governance Process (see 3.6 in Figure 7). Once dataset holders accept, the coordinating body recategorizes the draft rule inventory to a final rule inventory that is ready for harmonization (see 3.11-3.12 in Figure 7).

Once the rule inventory is final, the coordinating body maps governance rules across datasets (see 3.13 in Figure 7) in a tool like the rule harmonization matrix (see [Template 3](#), a suggested template for rule mapping). Using the common organization by governance domains and subdomains, the coordinating body compares rules across datasets and applies one of four categories:

- **Duplicate Rules:** Rules that are identical across datasets. These rules are deduplicated and one flows through unchanged to the consolidated rule set.
- **Unique Rules:** Rules that do not overlap with rules from other datasets. These rules flow through unchanged to the consolidated rule set.

- **Overlapping Rules:** Rules that are semantically different but materially the same: they may use different wording, phrasing, or other specifics but impose comparable requirements or achieve the same outcomes. For example, one dataset may require that data be stored in "secure, access-controlled environments," while another specifies that data must be stored in "restricted systems with role-based **access controls**." Although phrased differently, both rules describe the same underlying access/security requirement. Overlapping rules require harmonization to ensure clarity on investigator/requesting institution obligations while eliminating – where possible – duplicative and unclear requirements.
- **Conflicting Rules:** Rules that impose incompatible requirements that cannot be simultaneously satisfied. For example, one dataset may require data to be encrypted using a specific algorithm, while another may require a different encryption standard for compliance with its governance policies. Conflicting rules require harmonization to find a middle ground or an acceptable compromise that also clarifies investigator/requesting institution obligations and is agreeable across dataset holders.

Based on these categorizations, the coordinating body creates the consolidated rule set using the rule harmonization matrix template (Table 7). The duplicate and unique rules flow directly from the rule inventory into the consolidated rule set. For overlapping rules, the coordinating body may select one of the overlapping rules for the consolidated rule set or draft a new rule that may be acceptable to all dataset holders and serves as a middle ground, blending aspects of the rules that overlap. For conflicting rules, the coordinating body drafts a harmonized rule for the consolidated rule set that reflects the shared intent of the conflicting rules. Conflicting rules may require the coordinating body to consult the policies and governance sources to consider how much flexibility is available. Harmonized rules should balance the needs of all dataset holders and be grounded in the Linkage Governance Framework's guiding principles. When possible, the coordinating body may reference past rule harmonization activities where similar rule conflicts were overcome, and the resulting rules were documented in an accessible governance source.¹⁶

The coordinating body then organizes the governance rules in the consolidated rule set by parts of the data lifecycle (e.g., rules about data access), which marks the end of the coordinating body's independent effort to gather, analyze, and reconcile governance rules on behalf of the dataset holders (see 3.13 in Figure 7). This version of the consolidated rule set is a draft. These efforts provide a baseline for dataset holder cooperation and collaboration based on the coordinating body's trusted role and responsibility to approach rule harmonization in an unbiased way.

Upon completing the draft consolidated rule set, the coordinating body distributes the rule harmonization matrix (including the consolidated rule set) to the dataset holders for review and consideration. The rule harmonization matrix allows dataset holders to see how the rules from each dataset were aligned, categorized, and harmonized within the consolidated rule set. With this package available to inform their considerations, the dataset holders evaluate whether their organization can accept the draft consolidated rule set (see 3.14-3.17 in Figure 7). This is the third – and perhaps most critical – governance decision (see 3.17 in Figure 7).

If a dataset holder accepts the consolidated rule set, they notify the coordinating body. If a dataset holder cannot accept the consolidated rule set, they send their concerns to the

¹⁶ The centralized platform mentioned previously should collect this information.

coordinating body to assess whether a resolution is possible (see 3.18 in Figure 7). Dataset holders may also elect to explore exceptions that could allow the request to proceed. When resolvable, the coordinating body establishes a forum for dataset holders to collaboratively explore changes to the consolidated rule set (see 3.20 in Figure 7) through meetings with dataset holders, coordinated emails, or other formats for developing solutions that are agreeable to all. The goal is to identify where flexibility may allow the consolidated rule set to proceed. If a resolution can be reached, the coordinating body updates and redistributes the consolidated rule set for review by the dataset holder (see 3.14-3.17 in Figure 7). Once the coordinating body receives all dataset holder acceptances (see 3.21-3.22 in Figure 7), the coordinating body marks the consolidated rule set as accepted by the dataset holders (see 3.23-3.25 in Figure 7).

For some requests, a consolidated rule set that is acceptable to all dataset holders may not be possible. When the dataset holders cannot accept the consolidated rule set because outstanding conflicts cannot be resolved, these impasses represent critical learnings for the framework and the coordinating body. The coordinating body documents why the consolidated rule set could not be accepted, bypasses the fourth governance decision (Can the investigator/requesting institution comply with the consolidated rule set), and proceeds to the fifth and final governance decision, which is discussed below.

When the consolidated rule set has been accepted by all dataset holders, the coordinating body provides the accepted consolidated rule set to the investigator/requesting institution to review and confirm whether the study can comply with it – the fourth governance decision (see 3.26-3.30 in Figure 8). When confirmed by the investigator/requesting institution, the coordinating body finalizes the consolidated rule set. If the investigator/requesting institution cannot comply with the consolidated rule set, they provide feedback to the coordinating body (see 3.31-3.35 in Figure 8) and the process returns to dataset holders and further rule harmonization (see 3.18 in Figure 7). Alternatively, the investigator/requesting institution may decide to not pursue further harmonization and instead cancel the request, triggering alerts to the coordinating body and the dataset holders (see 3.36-3.50 in Figure 8).

When the investigator/requesting institution confirm that the study can comply with the consolidated rule set (see 3.30-3.43 in Figure 8), they send a confirmation to the coordinating body and the acceptance triggers the fifth and final decision where the dataset holder decides if the request is approved or denied (see 3.55 in Figure 9). The coordinating body notifies the dataset holders that the request is ready for decision and asks that they conduct one final request review and either approve or deny it¹⁷ (see 3.44-3.49 in Figure 8 and 3.51-3.55 in Figure 9). Without an accepted consolidated rule set, a denied request is expected. When the consolidated rule set was accepted and finalized, an approved request is expected. When all dataset holders issue their denial or approval to the coordinating body, the coordinating body labels the request as denied or approved and notifies the dataset holder and investigator/requesting institution (see 3.65-3.73 in Figure 9). If the request is not approved by one or more of the dataset holders, the coordinating body alerts the investigator/requesting institution so they may decide whether to pursue resubmission (see 3.56-3.64 in Figure 9). This concludes the Common Governance Process.

Following the completion of the Common Governance Process and an approval to move

¹⁷ It is unlikely that a request will reach the approval stage and not be approved by a dataset holder. Nonetheless, if a dataset holder does not approve the final consolidated rule set, the Coordinating Body is notified and determines the next steps for the request (see 3.43-3.46 in Figure 6).

forward, the dataset holders will next determine how to construct DUA(s) with the investigator/requesting institution.¹⁸ While this activity falls outside of the scope of the Common Governance Process, executing a DUA is an essential governance milestone. A signed and executed DUA is the governance output that allows data access, sharing, linkage, and use to begin.

4.4 Outcomes of the Common Governance Process

The Linkage Governance Framework is currently a proof of concept, testing the feasibility, practicality, and effectiveness of a collaborative process to significantly innovate governance for data linkage for research. The Common Governance Process has three potential outcomes: an approved request, denied request, or withdrawn request. For an approved request, dataset holders have reached agreement on how the datasets will be linked (i.e., the linkage implementation) and what rules will govern the linkage implementation and resulting linked dataset (i.e., the consolidated rule set) such that a DUA(s) can be created and the study may begin. For a denied request, the dataset holders have determined that unresolvable rule conflicts or technical barriers make it impossible for the proposed linkage to proceed and identify the rules that blocked the linkage and, in some instances, noted recommendation for changes that may result in approval. For a withdrawn request, the investigator/requesting institution has determined that technical barriers or governance requirements from one or more dataset holders make it impossible for the proposed linkage to proceed, usually based on feedback from the coordinating body or dataset holders.

¹⁸ The rationale for excluding DUA creation from the process is that 1) the coordinating body as the facilitator is not a signatory on the DUA, 2) the dataset holders and requesting institutions should determine the most efficient approach to executing a DUA(s) for the study, and 3) some organizations prefer to coordinate and negotiate DUAs in private, where the presence of the coordinating body could be a barrier.

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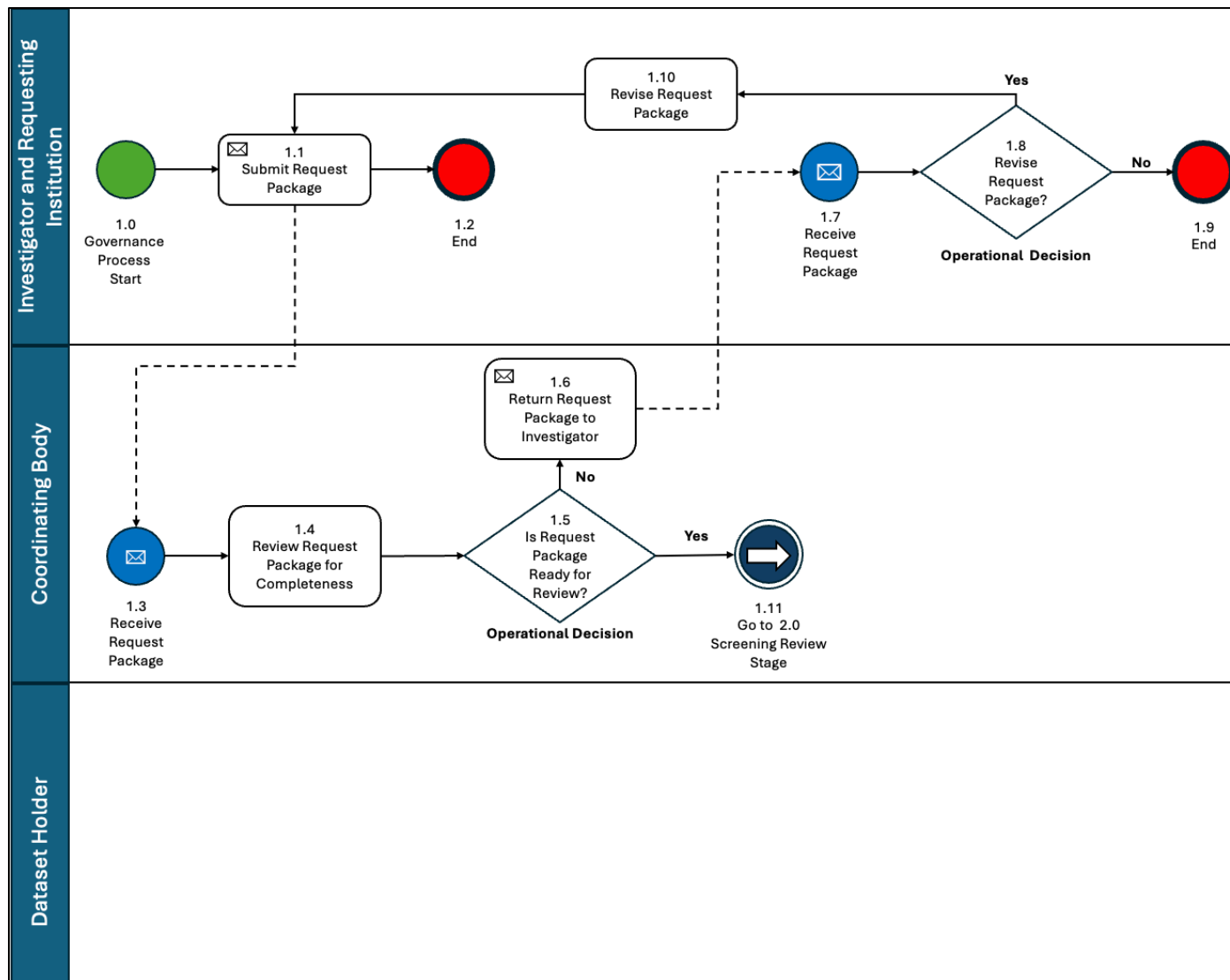


Figure 5: Intake Business Process Diagram

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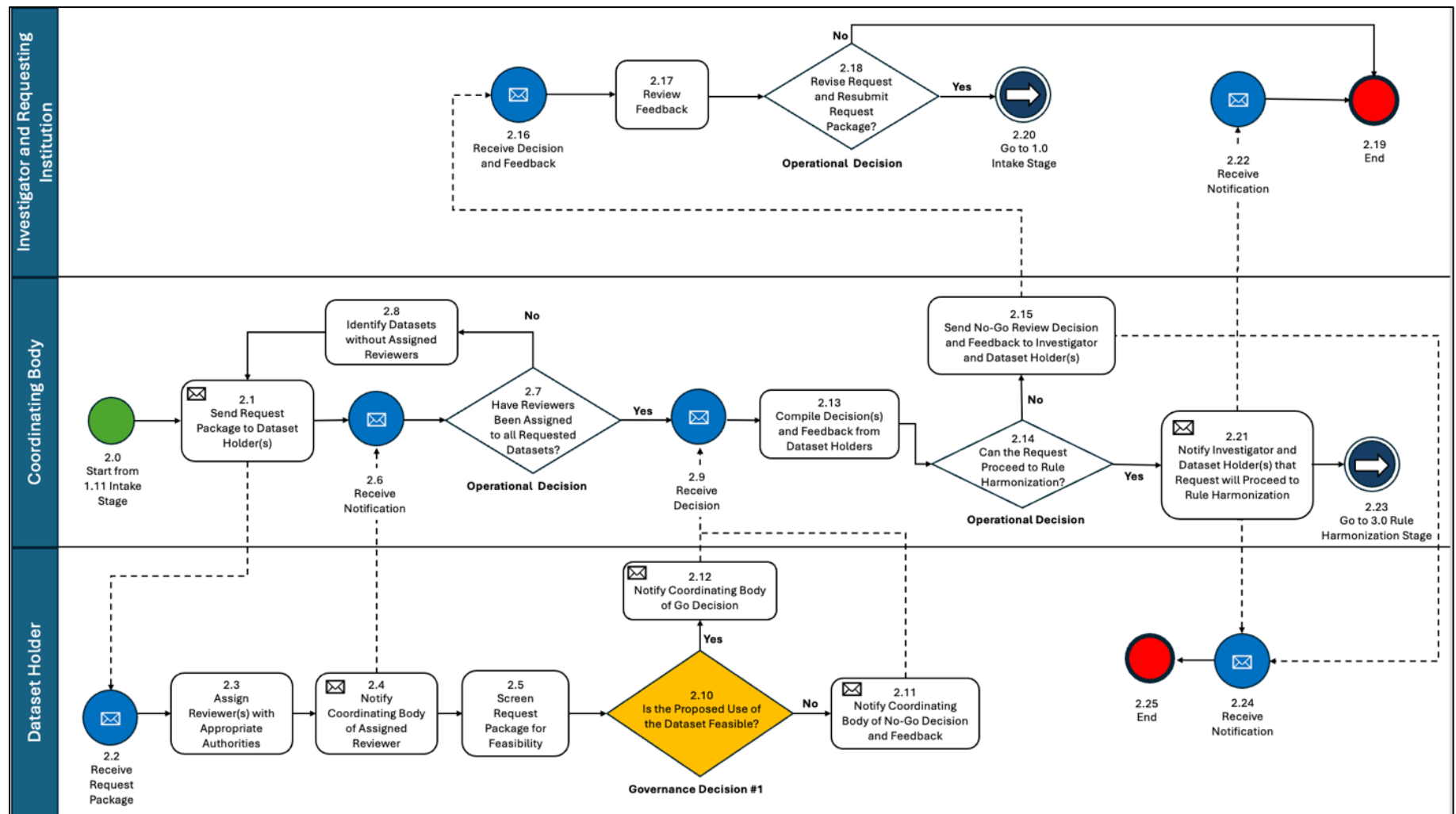


Figure 6: Screening Business Process Diagram

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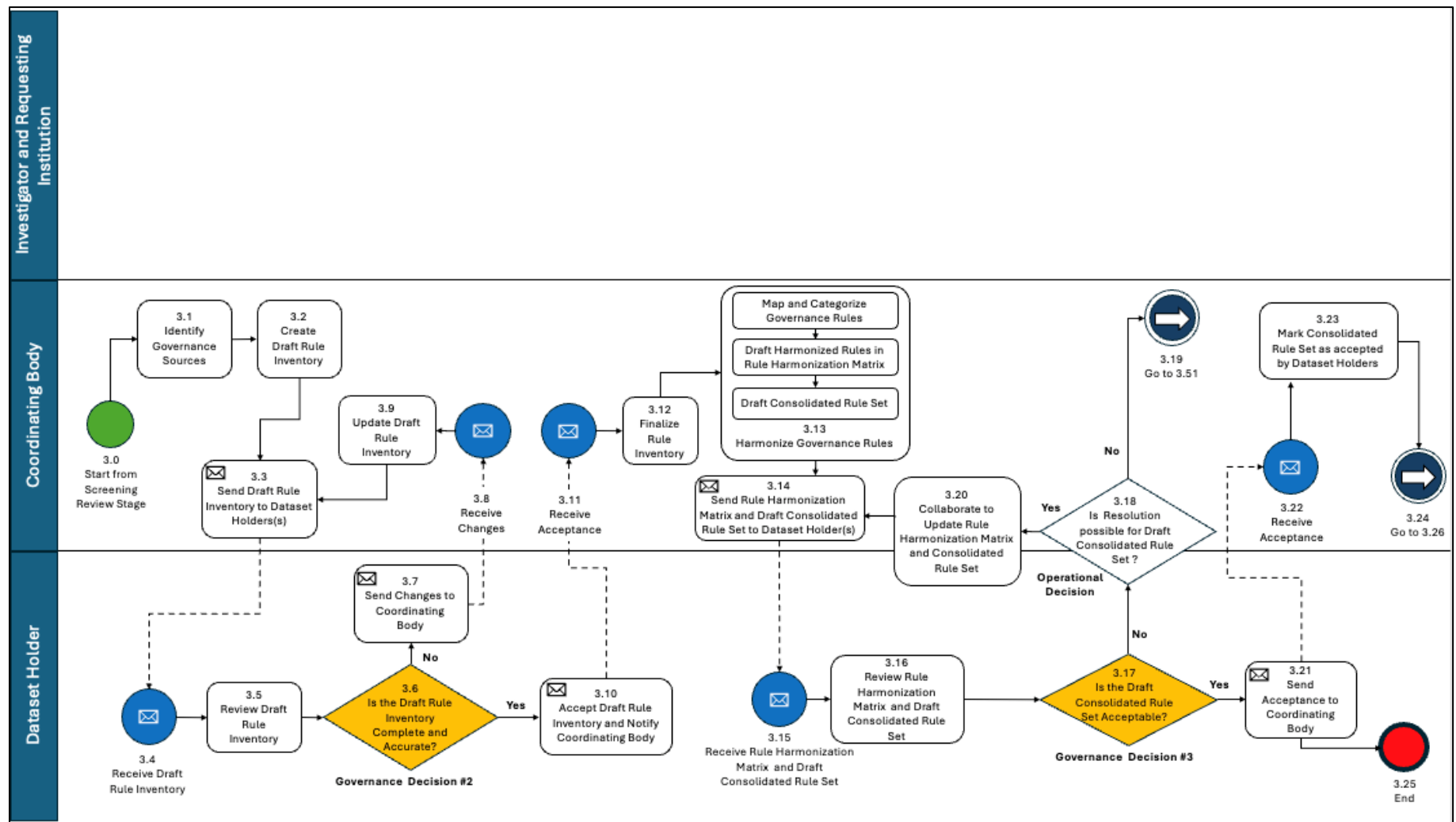


Figure 7: Rule Harmonization Business Process Diagram (1 of 3)

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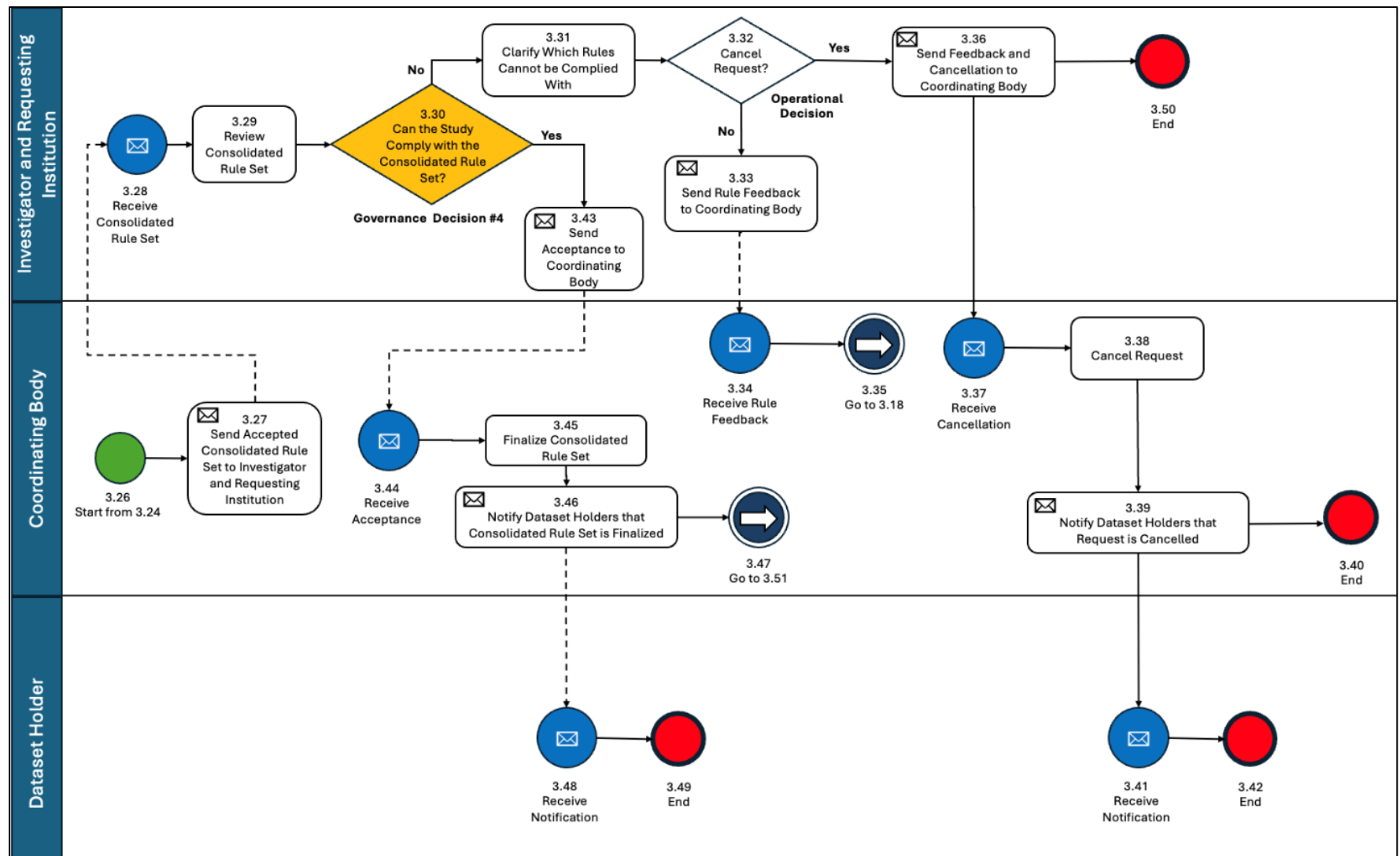


Figure 8: Rule Harmonization Business Process Diagram (2 of 3)

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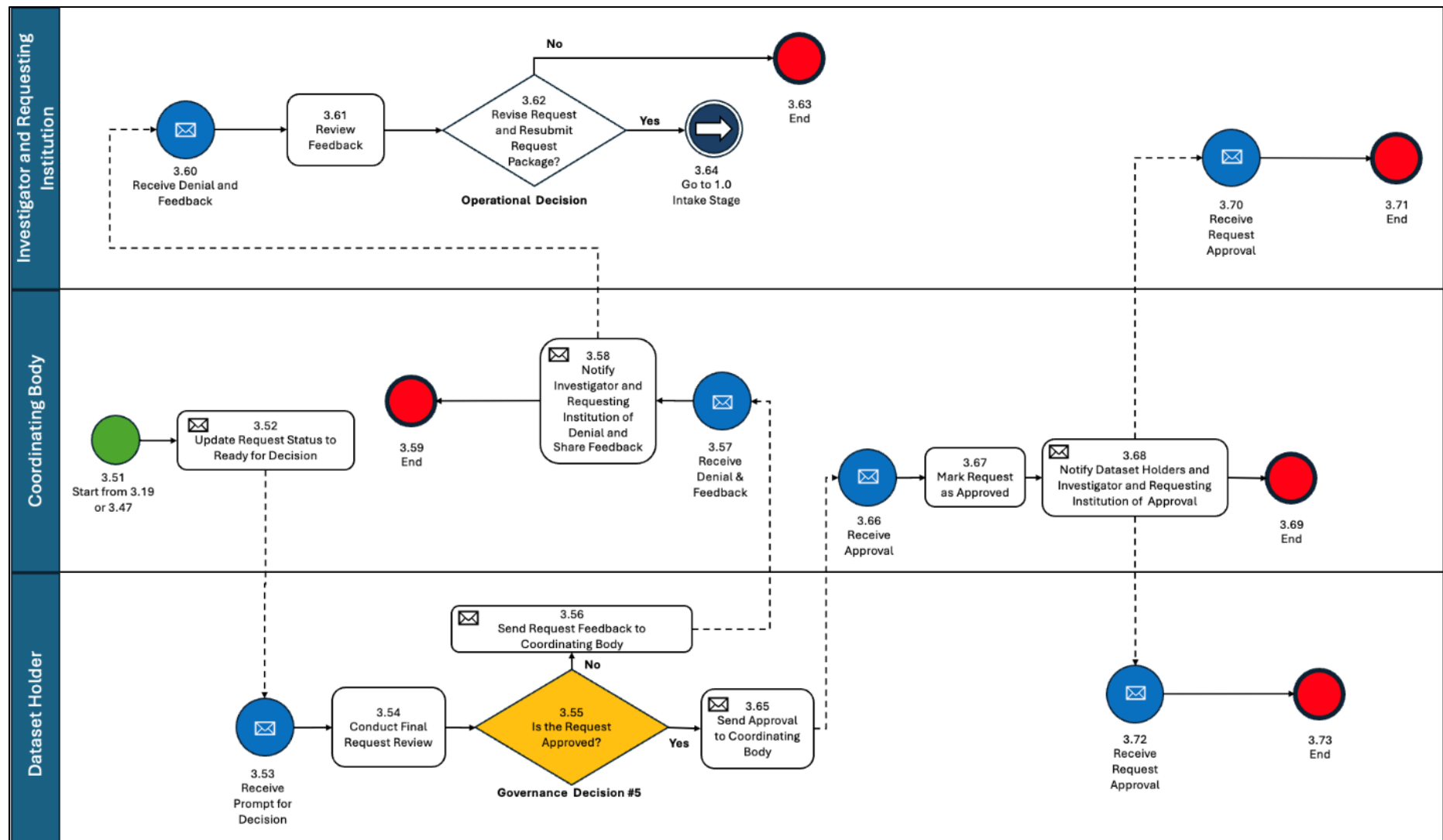


Figure 9: Rule Harmonization Business Process Diagram (3 of 3)

5. Roles and Responsibilities

As described in the Common Governance Process above (see Section 4), the Linkage Governance Framework recognizes four roles: investigator, requesting institution, dataset holder, and coordinating body. Each role has many responsibilities that play a part in the safe and secure linkage of individual-level health data for research and the governance activities that allow that research to occur. Building on the description of each role's responsibilities within the process in [Section 4](#), this section summarizes process responsibilities by role and describes additional responsibilities that fall outside the governance process but provide necessary context and are essential for the design, conduct, and completion of a study. Including these responsibilities helps illustrate how the governance process intersects with the research lifecycle and clarifies the expectations for each role within governance. It also makes clear which responsibilities do not fall under specific roles.

5.1 Investigator and Requesting Institution

An investigator is the individual who is responsible for the design, conduct, reporting, and scientific and ethical integrity of a research study. The investigator plays a critical role ensuring both the study's execution and the research team's actions are compliant with laws, regulations, and funder and institutional policies. A requesting institution is the entity with which the investigator is affiliated, the signatory of any governance agreements, and the entity ultimately responsible for the research study. Requesting institutions could include but are not limited to universities, laboratories, non-profit and for-profit entities, and government. Because research operations vary across requesting institutions, the framework combines the responsibilities of the investigator and requesting institution, indicated by "investigator/requesting institution," so that the institution is ultimately responsible for oversight of the investigator(s) but also enables the flexibility to delegate responsibilities that reflect the policies and practices of the requesting institution. Table 1 outlines the responsibilities of the investigator/requesting institution.

Before beginning a request package, the investigator/requesting institution are expected to have designed the study (potentially including engagement of patient and stakeholder perspectives as appropriate to the research and requesting institution's processes), defined an approach and methods to link data, and completed IRB and Privacy Board reviews and received approvals or provided other documentation indicating waivers, non-applicability of those governing bodies, etc. The investigator/requesting institution are also expected to be familiar with the Common Governance Process and the specific governance requirements for the datasets they are requesting.

Table 1: Investigator/Requesting Institution Responsibilities

Governance Process Stage	Investigator/Requesting Institution Responsibilities
Intake	Complete, submit, and revise the request package as needed.
Screening	Determine whether to revise the request package in the event of a "no-go" decision and revise request package as needed.
Rule Harmonization	Decide whether the study can comply with the consolidated rule set. Determine whether to revise and resubmit the request package, if denied.

Upon approval of the request and completion of the process, the investigator/requesting institution will work on several additional steps that are outside of this process like executing a DUA(s) with the dataset holders (see the end of [Section 4.3](#)). Additional contracts, DUAs, or other types of agreements may also be needed with other partners involved in conducting the linkage. The investigator and their team may need to complete additional trainings, make payments, establish access to specific platforms, or undertake other activities that are required by dataset holders before data may be accessed.

Once the data linkage(s) and research are underway, the investigator/requesting institution are accountable to abide by the consolidated rule set as well as report their progress conducting the linkage and carrying out the study to dataset holders via requirements set by the governance agreement. Governance agreements will also direct the investigator/requesting institution on what is a reportable event and how to report those events to the dataset holders.

5.2 Dataset Holder

The dataset holder is an organization with the authority to make decisions about the access, sharing, linkage, and use of a specific dataset. A **data owner** or **data steward** may fulfill this role; however, not all data owners or data stewards hold this authority. Dataset holders act as gatekeepers, balancing their obligations to protect sensitive personal information and uphold dataset policies with the goal of advancing research through responsible data sharing. They create and enforce policies for data access, sharing, linkage, and use and monitor users to ensure that data linkage has been conducted ethically, responsibly, transparently, and in accordance with applicable policies and protections. Participation in the Linkage Governance Framework is voluntary, and dataset holders retain full autonomy to decide whether to engage in it. This discretion allows organizations to align their participation with their missions, values, and operational capabilities. Dataset holders may name reviewers to participate in the governance process: a reviewer is an individual who has strong knowledge of the requested dataset and is able to make (or coordinate) timely recommendations on the request package during Screening and in the subsequent Rule Harmonization stage. Table 2 outlines the responsibilities of participating dataset holders during the governance process.

Table 2: Dataset Holder Responsibilities

Governance Process Stage	Dataset Holder Responsibilities
Intake	N/A
Screening	Review the request package and decide whether the proposed use of the dataset is feasible (a “go” or “no-go”) to proceed to rule harmonization.
Rule Harmonization	Provide governance sources. Review and provide feedback on the draft rule inventory and rule harmonization matrix. Work with the coordinating body and other dataset holders toward agreement on the consolidated rule set. Decide to approve or deny the request.

As with the investigator role, some study-specific governance activities are outside the scope of

the Common Governance Process and must occur after the Common Governance Process has been completed. For example, the dataset holder works with the requesting institution to develop and sign governance agreements (e.g., DUA[s], contracts, and other agreements) and works directly with the investigator and study team to ensure that all other requirements (e.g., trainings and collection of fees) are satisfied before granting access to data. The dataset holder also may have ongoing responsibilities for monitoring research and closeout to ensure compliance with the DUAs.

5.3 Coordinating Body

The coordinating body is a new role created by the Linkage Governance Framework with both strategic and operational governance responsibilities. As a neutral facilitator and operator of the Common Governance Process, the coordinating body creates efficiency in governance decision making, ensures transparency in the process, promotes collaboration, and enables responsible data linkage for research. The organizational structure and home of the coordinating body is yet to be defined. Required expertise includes research operations, HHS data ecosystem, linkage, and data governance and policy. The Coordinating Body could be overseen by the HHS agency responsible for the Linkage Governance Framework. Numerous approaches could be taken to establish and maintain a Coordinating Body including roles filled by a contract organization, by a small cadre of staff from the agency overseeing the framework, or by a mixture of the two.

At the strategic level, the coordinating body provides leadership, oversight, and direction to ensure the framework delivers value, aligns with HHS priorities, and evolves to meet the changing needs of the research data environment. At the operational level, it serves as the primary interface between dataset holders and the investigator/requesting institution during the Common Governance Process. The coordinating body also plays a crucial role in the framework's continuous improvement efforts (see [Section 8](#)), ensuring that the Linkage Governance Framework continues to evolve and improve over time. Through these efforts, the Coordinating Body advances research, fosters trust, and supports the evolving data linkage ecosystem. Table 3 outlines the coordinating body's governance process responsibilities.

Table 3: Coordinating Body Responsibilities

Governance Process Stage	Coordinating Body Responsibilities
Intake	Process the request package and coordinate with the investigator/requesting institution to ensure it is complete. Move the completed request to the Screening stage.
Screening	Facilitate dataset holders' review of the request package. Communicate with the investigator/requesting institution and dataset holders to gather further information and regarding Screening decision.
Rule Harmonization	Develop and revise the draft rule inventory, rule harmonization matrix including consolidated rule set. Coordinate dataset holders' review and revision of the consolidated rule set. Coordinate investigator/requesting institution's consolidated rule set review and feedback. Communicate with the investigator/requesting institution and dataset holders regarding final request decision.

6. Governance Decisions

The most important function of the Common Governance Process is to facilitate governance decisions between dataset holders and investigator/requesting institutions that ultimately result in permission to link the data, or a decision not to link. Five governance decisions are made during the Common Governance Process as well as a handful of operational decisions (see Figure 5-8). Operational decisions are important inflection points in process navigation that inform what happens next and ensure the process continues to completion. Governance decisions, which are the focus of this section, rely on policy and legal determinations. This section reviews the four decisions made by the dataset holder and one decision made by the investigator/requesting institution. Table 4 illustrates these five governance decisions and outlines the inputs and outputs for each decision. Each governance decision relies on inputs, such as documents or information, to support informed decision making and produces outputs that mark progress within a stage. This structured approach ensures accountability and clarity in the process.

Table 4: Governance Decision Matrix

Decision Maker	Screening Governance Decisions	Inputs	Outputs
Dataset Holder	1. Is the proposed use of the dataset feasible?	Request package	“Go” or “no-go” decision to Rule Harmonization stage
Dataset Holder	2. Is the draft rule inventory complete and accurate?	- Rule Harmonization Protocol - Draft rule inventory	- Suggested revisions to the draft rule inventory - Final rule inventory
Dataset Holder	3. Is the draft consolidated rule set acceptable?	- Draft rule harmonization matrix - Draft consolidated rule set	- Suggested revisions to the consolidated rule set - Accepted consolidated rule set
Investigator/Requesting Institution	4. Can the study comply with the consolidated rule set?	Accepted consolidated rule set	- Challenges to compliance with the accepted consolidated rule set - Final consolidated rule set
Dataset Holder	5. Is the request approved?	- Request package - Final consolidated rule set	Formal request approval or denial

7. Implementation Tools

The Linkage Governance Framework requires resources to support its adoption and future implementation. This proof-of-concept framework initially includes three implementation tools (described below) that explain specific aspects of the framework and Common Governance Process in greater detail: Standards Supporting the Common Governance Processes, Request Package Overview, and Rule Harmonization Protocol. The tools described below are not a complete set of implementation materials and are in draft form. Each of these three tools will evolve along with the framework. More implementation tools will be developed as the framework evolves in its implementation readiness. Playbook sections above have referenced these tools when relevant, and how and when to use each tool is described below.

7.1 Standards Supporting the Common Governance Processes

The Standards Supporting the Common Governance Processes ([Appendix C](#)) is a set of 12 standards created to translate the guiding principles ([Section 3.4](#)) to a process and operational level. They establish a clear and consistent direction for the conduct of governance activities in the Common Governance Process. The principles and standards together serve the purpose of a traditional “policy document” that governing bodies often use to formalize the rules and procedures that guide governance decision making. The standards translate the core values established in the guiding principles into specific, operational requirements that clarify roles and responsibilities, foster transparency, preserve dataset-specific policies, and ensure a balance between risk and research benefit.

7.2 Request Package Overview

The Request Package Overview ([Appendix D](#)) describes the information that an investigator/requesting institution will compile and submit to request access to individual-level health datasets for linkage. It is the first step in the Common Governance Process. The overview also describes the steps involved in compiling and submitting the request package, as well as how the information submitted may be used. Functionally, the Request Package Overview details the recommended elements that could inform the design of a future request tool. Investigators/Requesting Institutions could use such a tool to understand and consider the work to create, complete, and submit a request package for a research study involving linkage. dataset holders may also use a tool like this to understand what information will be provided by an investigator/requesting institution and to anticipate the details they will receive to facilitate decision making.

7.3 Rule Harmonization Protocol

The Rule Harmonization Protocol ([Appendix E](#)) supports the coordinating body in facilitating the rule harmonization process. It outlines a structured protocol for rule harmonization, a process designed to align and integrate governance rules across multiple datasets and facilitate linkage decision(s). The protocol describes how to compare governance rules across datasets, categorize groups of similar rules, resolve overlapping or conflicting rules, and generate a single consolidated rule set.

8. Continuous Improvement

The Linkage Governance Framework includes a continuous improvement component that aims to ensure the framework adapts to the changing needs of the data linkage and research ecosystem. Key opportunities for the Linkage Governance Framework include ongoing learning from:

- The experience of the framework and Common Governance Process participants,
- The outcomes of submitted requests, and
- The successes and challenges of linkages and associated research studies that result from approved requests.

Continuous improvement activities for this framework may include:

- Defining performance metrics and beginning to track performance data accordingly,
- Collecting information on the above topics and compiling that information in a structured fashion,
- Regularly reviewing the information collected to identify themes and patterns where the framework's performance could be improved, and
- Implementing changes in the framework to address inefficiencies or suboptimal performance.

The coordinating body is primarily responsible for conducting and coordinating continuous improvement. This work should occur independent of the Common Governance Process, under the oversight of its home agency or another HHS entity. One approach to ensuring continuous improvement occurs as intended is to incorporate a requirement to participate into the DUA that binds investigator/requesting institution and dataset holders. Toward this end, the potential continuous improvement workstreams are:

1. **Documenting each linkage request with detailed metadata:** The coordinating body, through a supporting technology platform, documents how each linkage request progresses through the Common Governance Process. This begins with the request record initiated when the request package is submitted. Throughout this process, the coordinating body records the decision makers involved, the decisions made, and any feedback provided. As the Common Governance Process is used, a secondary asset is created: a library of linkage governance information. By marrying **metadata** about a request, its review, resulting decisions, and the ultimate outcome (approval, denial, or cancelation), the coordinating body, dataset holders, and the investigator/requesting institution may be able to:
 - Search for requests of similar studies and their outcomes,
 - Identify examples such as linkage implementations that were approved to reuse and guide requests, and
 - Learn from whether and how conflicting governance rules were ultimately harmonized.

2. **Collecting user feedback about the experience of the Common Governance Process:** After a request has proceeded through the Common Governance Process, the coordinating body may contact the participating dataset holders and investigator/requesting institution to gather feedback on their experience including things that worked well and things that did not, as well as recommendations to optimize workflows and improve implementation tools, reduce level of effort, and improve user experience.
3. **Linkage documentation:** For approved requests, the coordinating body may work with the dataset holder and/or investigator/requesting institution to track the outcome of the linkage implementation. This may include recording in the request record whether the linkage was successful and capturing reflections on how the linkage implementation could have been improved. By systematically connecting linkage outcomes with the proposed implementation and accumulating this information over time, the resulting library of linkage information could help distill which implementation designs and features are most likely to lead to successful linkages. Additionally, each successful linkage serves as evidence of a linked dataset's existence. In this way, linkage outcomes may also act as an index of linked datasets, and the additional metadata about what led to that dataset's creation provides the provenance and approved use of that linked dataset.
4. **Closeout:** The coordinating body may retrieve and document the outcomes of the studies completed from approved requests, connecting the request record with the resulting publications and the disposition or dissemination of the linked dataset. In this activity, the coordinating body should be careful to not create duplicate efforts for the investigator/requesting institution, recognizing that dataset holders may already require investigators to report study outcomes, research findings, and resulting publications.
5. **Evaluating the coordinating body's role and performance:** The coordinating body's role as the operator of the Common Governance Process is important to evaluate. To support continuous improvement, the coordinating body or other external party/entity may conduct retrospective reviews of its role in processing requests and engaging with dataset holders and investigators/requesting institutions. Dataset holders and requesting institutions may be asked to provide feedback about their experience interacting with the coordinating body.

Appendix A. Glossary

This glossary defines certain terminology used in Linkage Governance Framework materials to foster a shared understanding of the language and concepts that underpin the framework, and to facilitate clear communication and consistent application. It includes common terminology that has been tailored to the specific context and concepts relevant to the framework, as well as new terminology developed for the framework to describe new roles and concepts specific to the framework.

Access Controls: Mechanisms or policies that restrict and regulate who can access data, under what conditions, and through what systems. Examples include role-based access controls, multi-factor authentication, and secure environments.

Linkage Governance Framework: A structured approach for seeking approval from dataset holders and harmonizing governance rules across datasets to enable safe and secure linkage of individual-level health data for research.

Complex Linkages: Linkage of individual-level health data for research where establishing the linkages entails a high level of governance complexity.

Consolidated Rule Set: A single set of rules that applies to a linkage implementation and the resulting linked dataset and reflects the governance rules from the originating datasets.

Coordinating Body: The entity that operates the Linkage Governance Framework with both strategic and operational governance responsibilities to lead the Common Governance Process.

Data Access, Sharing, Linkage, Use, and Dissemination

- **Access:** The processes and controls that determine who can view or use data, under what circumstances, and by what means.
- **Sharing:** The process of providing access to data to authorized parties for approved purposes like research or collaboration.
- **Linkage:** The process of combining individual-level health data from two or more datasets at a person level to create a single linked dataset.
- **Use:** The application of data for a specific purpose, such as research, in accordance with its governance.
- **Dissemination:** The process of distributing data used in or created from research – in its original or derivative form – to authorized parties or the public, governed by rules that specify recipients, conditions, safeguards, and restrictions. Distinct from dissemination of research findings.

Data Collection: The process of obtaining data from participants, systems, or other sources for research, clinical, or administrative purposes.

Data Governance: The policies, processes, and resulting decisions that specify how data can be accessed, shared, linked, used, and disseminated.

Data Lifecycle: The process of accessing, sharing, linking, using, and disseminating data that is fundamental to research with linked data.

Data Owner: The person or entity that has legal control of and responsibility for data.

Data Steward: A formal position or an assigned accountability from the Data Owner for a specific dataset and the processes that surround it.

Data Use Agreement (DUA): A formal agreement between parties that defines the terms and conditions for accessing, sharing, linking, using and disseminating a dataset.

Dataset: An organized collection of data.¹⁹

Dataset Holder: A Data Owner or Data Steward responsible for access to and use of a dataset and compliance with applicable governance.

De-identified Data: Data for which elements of protected health information and personally identifiable information like names, dates of birth, and phone numbers are removed in order to mitigate privacy risks to individuals.²⁰

Governance: A framework for decision making and clarifying roles and responsibilities based on established policies and processes.

Governance Rule: A specific directive or guideline that dictates how data is managed, accessed, shared, linked, used, or disseminated within an organization or dataset system or between organizations.

Human Subjects Research: Research involving a living individual about whom data or biospecimens are obtained/used/studied/analyzed through interaction/intervention, or identifiable, private information is used/studied/analyzed/generated.²¹

Individual-level Health Data: Health-related person-level data that is found in electronic health records (EHRs), research repositories, registries, and other data systems. May or may not include protected health information or personally identifiable information.

Institutional Review Board (IRB): A formally designated group that helps protect research participants from research-related risks and ensure the ethical conduct of research by reviewing research studies to ensure that they comply with applicable regulations, meet commonly accepted ethical standards, follow institutional policies, and adequately protect research participants. An IRB has the authority to approve, require modifications in (to secure approval), or disapprove research.

Investigator: The individual with overall responsibility for the design, conduct, reporting, and scientific integrity of a research study.

Linkage Implementation: A detailed plan describing how datasets will be linked, including

¹⁹ Source: <https://resources.data.gov/glossary/dataset/>

²⁰ Source: <https://www.hhs.gov/hipaa/for-professionals/special-topics/de-identification/index.html#rationale>

²¹ Source: <https://grants.nih.gov/policy-and-compliance/policy-topics/human-subjects/research>

access, transmission, sharing, person matching, data privacy, and security.

Linked Dataset: A dataset created by merging information about the same person from separate, originating datasets into one unified dataset.

Metadata: Data that describes data. For example, metadata may include the date of collection, ranges of acceptable values, and relationships to other data. Metadata may also include governance information such as the responsible steward, associated laws and regulations, and access management policy.

Originating Dataset: A dataset that will be linked at the individual level and will provide data attributes to a linked dataset.

Personally Identifiable Information (PII): information that can be used to distinguish or trace an individual's identity, either alone or when combined with other personal or identifying information that is linked or linkable to a specific individual.²²

Privacy Board: A formally designated group that reviews research studies to ensure compliance with privacy laws and regulations. These functions may be consolidated under an IRB.

Protected Health Information (PHI): Individually identifiable health information held or transmitted by a covered entity or its business associate, in any form or media, whether electronic, paper, or oral.²³ Governed by the Health Insurance Portability and Accountability Act (HIPAA), PHI relates to an individual's health condition, healthcare provision, or payment for healthcare.

Requesting Institution: Any organization that requests access to data for linkage and use in research through the Common Governance Process, such as an institution of higher education, nonprofit or for-profit organization, local or federal government agency, etc.

Rule Harmonization: A structured process to align and reconcile governance rules across multiple datasets and create a single consolidated rule set for managing a linked dataset.

²² Source: <https://toolkit.ncats.nih.gov/glossary/personally-identifiable-information/> & <https://grants.nih.gov/grants/glossary.htm#P>

²³ Source: <https://www.hhs.gov/hipaa/for-professionals/privacy/laws-regulations/index.html#what>

Appendix B. Acronyms

Acronym	Definition
AI	Artificial intelligence
DUA	Data use agreement
EHR	Electronic health record
FFRDC	Federally Funded Research and Development Center
Health FFRDC	CMS Alliance to Modernize Healthcare
HHS	Department of Health and Human Services
HIPAA	Health Insurance Portability and Accountability Act
IRB	Institutional Review Board
LLM	Large language model
NCI	National Cancer Institute
NIH	National Institutes of Health
NIST	National Institute of Standards and Technology
OS-PCORTF	HHS Office of the Secretary Patient-Centered Outcomes Research Trust Fund

Appendix C. Standards Supporting the Common Governance Process

The Common Governance Process includes three stages (Intake, Screening, and Rule Harmonization) that together seek creative solutions to enable data linkage for research. The standards defined below explain how the guiding principles ([Section 3.4](#)) may be operationalized in the Common Governance Process ([Section 4](#)).

Standard 1: The Common Governance Process involves Intake, Screening, and Rule Harmonization, and results in the approval or denial of a requested data linkage.

There is a range of governance work that happens across the research lifecycle though not all of it is encompassed in this process. The Common Governance Process ([Section 4](#)) is limited to receiving and processing a request, facilitating screening by dataset holders, conducting rule harmonization to create a consolidated rule set, and coordinating the linkage request approval or denial. The process starts when a request package is submitted and ends when a request is approved or denied. The Common Governance Process does not engage in the governance activities that precede the request submission (e.g., scientific merit review, IRB and Privacy Board review, linkage design) and expects some governance activities to follow a linkage request approval (e.g., executing data use agreements or DUAs, executing contracts with business associates, collecting fees, implementing linkage).

Standard 2: The Common Governance Process is designed for investigators proposing complex data linkages for research.

The governance complexity surrounding linking datasets can delay or even derail research and some linkages are more complex than others. Complex linkage refers to linkages that entail a high level of governance complexity, including linking three or more datasets, thereby involving multiple sets of request forms, asynchronous review processes, complicated technical architecture, or a novel linkage approach. The creation of the Linkage Governance Framework was inspired by a desire to streamline complex linkages through a flexible and scalable governance process.

Standard 3: The Common Governance Process facilitates decision making through cooperation and collaboration.

The Linkage Governance Framework creates an interface between dataset holders and the investigator/requesting institution to enable decisions via the Common Governance Process with support from the coordinating body. The Framework does not establish a governing body. It facilitates decision making but does not make decisions. Dataset holders retain their authority to approve or deny requests to access and link data and approve or deny the consolidated rule set that will govern the resulting linkage implementation, research, and linked dataset. Investigators/Requesting Institutions are not beholden to the coordinating body or the Framework, rather they remain accountable to the dataset holders from which they request and receive data for linkage.

Standard 4: The Common Governance Process relies on clearly defined roles and responsibilities.

Clearly defined roles and responsibilities are essential for the Common Governance Process to operate effectively. The roles that participate in the Common Governance Process are the investigator/requesting institution, dataset holder, and coordinating body (see [Section 5](#)). All aspects of the process must identify the entity responsible for each activity and making each operational or governance decision.

Standard 5: The Common Governance Process prioritizes transparency to investigators/requesting institutions and dataset holders.

The Common Governance Process is intended to be transparent through clear and comprehensive process documentation, progress tracking by the coordinating body, and timely communication with the investigator/requesting institution and dataset holders. All parties will understand how the process works, what decisions need to be made, and where the request is within this process.

Standard 6: Participation in the Common Governance Process is voluntary.

The Linkage Governance Framework is not a mandate. Dataset holders and Investigators/Requesting Institutions determine if, how, and how much they will participate in the Linkage Governance Framework and may change their decision. Investigators/Requesting Institutions choose if they will submit their data linkage requests to the Common Governance Process or instead use existing governance processes. Dataset Holders choose which datasets are available through the Common Governance Process. While investigators/requesting institutions and Dataset Holders decide whether to engage with the process, the framework relies on widespread adoption and participation to achieve its ambitious objectives.

Standard 7: The Common Governance Process accommodates each dataset's unique governance and does not change Dataset Holders' policies.

The Common Governance Process is flexible and can accept any type of policy and the associated governance rules a dataset follows. The Common Governance Process harmonizes governance rules across datasets without altering dataset holder policies. When harmonized rules are not possible, the coordinating body works with dataset holders to determine what changes could make harmonization possible (see [Section 4.3](#)).

Standard 8: The Common Governance Process preserves Dataset Holders' responsibility to appraise risks associated with linking data.

The Linkage Governance Framework recognizes that permitting access and linkage of datasets is a decision that includes inherent risks for dataset holders. Granting an investigator/requesting institution access to a dataset increases a dataset's exposure to misuse, data security events, and privacy incidents involving individuals in that dataset. The Common Governance Process is premised on balancing risks with the potential benefits of the proposed research and supports dataset holders' careful review of risk(s) that may ultimately result in denying an investigator/requesting institution's request.

Standard 9: The Common Governance Process accommodates any safe and secure data linkage method.

Two common methods to link datasets used in biomedical research are clear text linkage²⁴ and privacy-preserving record linkage²⁵ (PPRL). The Common Governance Process is agnostic to linkage method and accepts requests regardless of method. Recognizing that some dataset holders have policies that limit linkage to one method, the Common Governance Process provides a forum where dataset holders may consider if and how alternate linkage methods can be made permissible.

Standard 10: A consolidated rule set must contain rules on critical linkage topics.

The consolidated rule set provides a single set of rules to support safe and secure linkage that, if followed, enables compliance with the policies of the requested datasets and dataset holders. While dataset holders will contribute a multitude of rules on various topics through rule harmonization, linkage-specific issues might not be addressed by current governance policies and could be poorly represented in the consolidated rule set as a result. Thus, the coordinating body and dataset holders must ensure that the consolidated rule set addresses the following linkage-specific topics:

- Permitted linkage method(s);
- Permissibility or prohibition of individual reidentification;
- Potential increased risk of reidentification of individuals due to linking datasets;
- Presence of **protected health information** (PHI) in the linked dataset;
- Ownership and stewardship of the linked dataset;
- Reuse and/or disposition of the linked dataset;
- How requested datasets will remain secure throughout the linkage process;
- Parameters for managing the linked datasets;
- Attribution of the dataset holders, requested datasets, and framework;
- Enforcement of governance rules throughout the research lifecycle; and
- Requirements for reporting the outcome of linkage, research results, and study completion.

Standard 11: The finalized consolidated rule set includes the rules that the requesting institution and the dataset holder agree can govern the linkage implementation and resulting linked dataset.

The consolidated rule set that is produced during rule harmonization is an essential output of the Common Governance Process (see [Section 4.3](#)). A consolidated rule set cannot be finalized until it has been approved by all dataset holders and accepted by the investigator/requesting institution. While this step concludes the formal Common Governance Process, the finalized

²⁴ Clear text linkage refers to the process of linking health data records that contain unencrypted personally identifiable information, such as names and dates of birth, by directly comparing these identifiers across different datasets to determine if records belong to the same individual.

²⁵ PPRL is “a means of connecting records using secure, pseudonymization processes in a data set that refer to the same individual across different data sources while maintaining the individuals’ privacy.” (Source: <https://covid.cd2h.org/PPRL/>)

consolidated rule set could be formally established for the proposed research by incorporating those rules into data use agreements between the investigator/requesting institution and the dataset holders.

Standard 12: Continuous improvement requires the Common Governance Process to evolve based on changes in and best practices from the data and research ecosystem.

Continuous improvement in the Common Governance Process includes soliciting input from partners to improve user experience, studying patterns in approved and denied requests, and identifying governance rules that block linkages from being approved (see [Section 8](#)). The coordinating body engages investigators/requesting institutions and dataset holders to learn about their experience and reveal how the process may be refined. Because denied requests can offer lessons about what causes denials and where in the process denials occur, the coordinating body reviews denial patterns to explore how the process can better support requests in achieving approval. The coordinating body records which rules are impossible to harmonize and passes back policy collisions to dataset holders to encourage alignment across the ecosystem.

Appendix D. Request Package Overview

Introduction

A request package is a collection of information that an investigator/requesting institution compiles and submits as part of the Common Governance Process to request access to multiple datasets with the purpose of linking and use in a research study. The request package is a critical input to the Common Governance Process' Intake stage ([Section 4.1](#)) and is a primary information source for dataset holders to decide if their dataset will be accessed, linked, shared, and used in the proposed study.

Following an overview of the request package, this appendix describes how an investigator/requesting institution prepares for submission, information required by the request package, the steps for submitting the package, and considerations for successful request package submission.

The primary audience for this overview is the investigator/requesting institution engaging in the Common Governance Process to seek approval to access and link requested datasets for research. A secondary audience is dataset holders participating in the Common Governance Process and receiving Request Packages for review and decision.

Request Package Overview

1. What Is a Request Package?

A request package is an application for dataset access specific to the Common Governance Process. An investigator/requesting institution creates and submits a request package to apply for access to datasets for the purpose of linking for a specific research study. Through the request package submission, the investigator/requesting institution aims to initiate a request that begins the Common Governance Process and provides dataset holders with the information needed to decide whether the requested dataset(s) can be linked and accessed for use in a research study.

2. Why Is a Request Package Necessary?

While linking individual-level datasets unlocks the potential to advance scientific discoveries, the governance complexity surrounding linking datasets can delay or even derail research. Defining a common request package creates an entry point for a unified process to request access to multiple datasets and builds consensus on what information is required for dataset holders to make decisions about linkage. Currently, the investigator/requesting institution must go through a process to request access for each dataset they hope to link. This means completing multiple request forms and multiple governance processes in parallel, each with distinct requirements, approval procedures, and timelines. If a change to the study design or linkage implementation is required based on feedback from one dataset holder, that change must be replicated with other dataset holders through other processes already under consideration. Engaging in multiple independent request processes is inefficient and can delay obtaining access to the data that investigators need to conduct their research.

Having a standardized request package as part of the Common Governance Process provides

opportunities to streamline, harmonize, increase efficiencies, and provide transparency and consistency for decision making.

3. Who Prepares and Submits the Request Package?

The investigator/requesting institution prepares the request package. The investigator may elect to prepare the request package themselves or choose to involve other members of their study team, such as the co-investigator(s). Norms at the Requesting Institution will ultimately determine who creates, reviews, and submits the request package.

4. How Is the Request Package Used?

The request package will be used in several ways. Submitting the request package begins the Intake stage, the first of three stages in the Common Governance Process. Submitting the Request Package creates a record for the study in the Common Governance Process wherein decisions about the request can be recorded, tracked, and made visible to the investigator, requesting institution, and dataset holders.

Upon receipt, the coordinating body will review the request package for completeness, ensuring it contains all required information with a sufficient level of detail. The purpose of reviewing for completeness is to assess whether the request package is ready for a dataset holder review, with the goal being that dataset holders do not have to go back to the investigator/requesting institution to ask for any additional information or clarification during their review.

Finally, the request package is what a dataset holder will review during the Screening stage to decide whether the proposed study and linkage are feasible. If determined to be feasible, the request will proceed to the Rule Harmonization stage. If governance rules can be harmonized and agreed on by the dataset holders and the investigator/requesting institution, the request will be approved.

Preparing to Submit a Request Package

In preparation for submitting a request package, the investigator/requesting institution is expected to have already designed the study, defined an approach and methods to link data, and received necessary approvals from their IRB and Privacy Board. This work is required ahead of the request package process because information from all three of these activities generates required parts of the request package.

1. Design the Study

Designing the study establishes the purpose, scope, and methodology for accessing and linking datasets. The request package assumes that the investigator has defined the research questions, hypotheses, and objectives, as well as specifying datasets that they desire to link (requested datasets), the linked dataset that will be created, analyses that will be conducted, and outcomes that will be assessed. The study design includes a justification for why the requested datasets are required for the research and defines the scope of the data for each requested dataset. As part of selecting requested datasets, the request package assumes that the investigator/requesting institution have conducted adequate discovery to understand the contents of the requested datasets, confirmed the availability of requested data elements, and familiarized themselves with each dataset's governance policies including permitted uses.

2. Plan the Linkage Implementation

The request package requires that the investigator/requesting institution have designed a plan for how requested datasets will be linked (a linkage implementation plan), which includes what organization will match individuals, what linkage method will be used, what software(s) will perform the linkage work, who will generate the linked dataset, and how data will be disposed of. As linkage includes accessing and transmitting multiple requested datasets to a common location for matching, a linkage implementation will specify data flow, privacy, and security throughout the process.

3. Secure Approvals

Before submitting the request package, the proposed study should have already been reviewed for ethical and privacy concerns based on the direction of the requesting institution. The request package expects that IRB and Privacy Board approvals or exemptions have been received prior to initiating the Common Governance Process. Formal documentation of study approvals or exemption is required, ensuring that governance work does not start for requests that may pose unacceptable ethical, human subjects use, and/or privacy risks.

Request Package Steps

Once preparation work is finished, the investigator/requesting institution can begin completing a request package. For a request package to be complete, an investigator/requesting institution must: 1) initiate a request; 2) provide administrative and study documents (such as IRB and Privacy Board approval letters); 3) provide information on a variety of study, linkage, and governance topics into the request form; 4) respond to any governance questions specific to the datasets that are being requested; 5) complete required attestations; and 6) affirm the request package is ready for submission. Figure 10 illustrates the six request package steps that address these requirements. Each step is explained below.

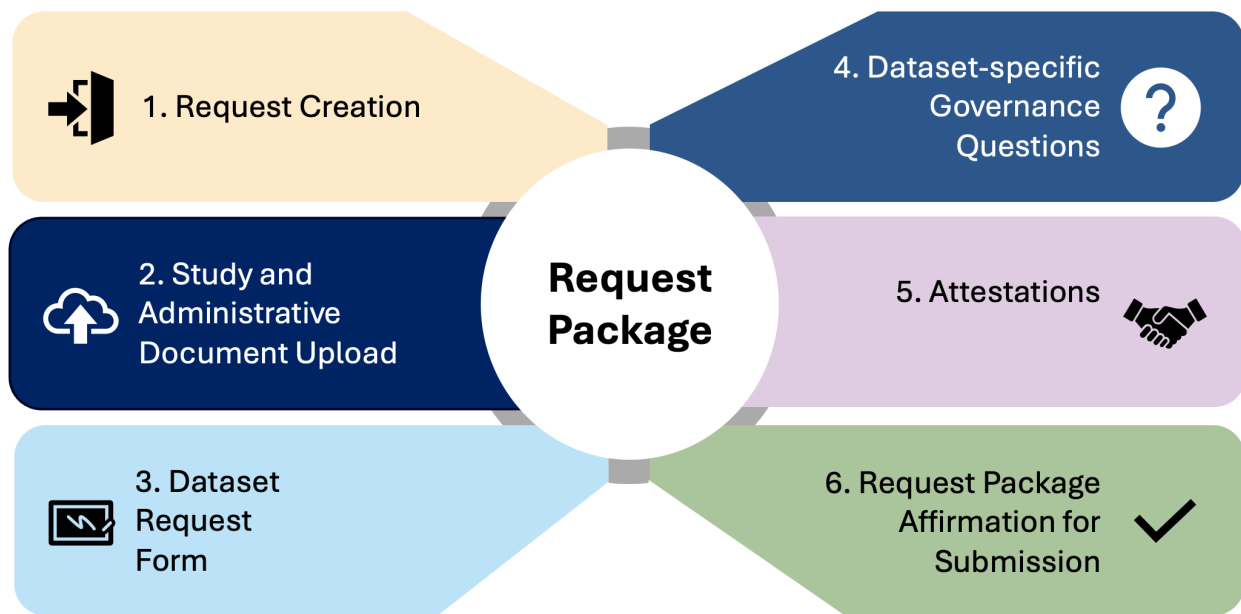


Figure 10: Request Package Steps

Recognizing that every study is unique and every investigator/requesting institution may organize their work differently, the steps described below do not have a prescribed sequence.

1. Request Creation

The investigator/requesting institution initiates a request by creating a new request package that is specific to the study they are proposing; the newly created request package is affiliated with the investigator and the requesting institution. Creating a request involves providing the title of the request (which can be the same as the study title), the Investigator's name and contact information, and the name and contact information for the Requesting institution. This action is always the first step as none of the other steps can be completed until a request package is created.

2. Study and Administrative Documents

The investigator/requesting institution provides study and administrative documents for the request package. For every request package, required documents include study description/protocols, submission materials, and approvals from the IRB and Privacy Board review. Additional documents may include any number of supporting materials and documentation including but not limited to consent forms, a determination that the work does not involve human subjects, a waiver of consent requirement, letters of support from dataset holders or other study participating organizations, training certificates, grant and award materials, relevant publications, and technical specifications. Some additional documents may also reflect dataset-specific required documentation.

Dataset holders review IRB and Privacy Board approvals to ensure that the IRB and Privacy Board have sufficiently examined the ethics of the study and risk of privacy, and that necessary approvals are in place. Dataset holders may use additional documents provided to ensure that specific governance requirements have been addressed or to gain more context about the study, linkage, or participating organizations.

3. Dataset Request Form

The dataset request form collects information on a variety of topics including but not limited to the study team, IRB and Privacy Board of record, requested datasets, linkage implementation, and privacy and security from the investigator/requesting institution. Each of the topics included in this form is detailed [below](#). Dataset holders use form responses to understand how the investigator/requesting institution plans to access, share, manage, link, and use the requested dataset(s) and linked dataset in the proposed study. One of this form's most critical functions is to identify the requested datasets for linkage that generate the dataset-specific governance questions.

4. Dataset-Specific Governance Question

The dataset-specific governance questions are questions about governance tasks required by the requested datasets (identified in the dataset request form). Many dataset holders require the investigator/requesting institution to complete governance tasks prior to requesting access to a dataset. Thus, dataset holders will want to confirm that required governance tasks have been completed when commencing their review. For example, one dataset may require that an

investigator establish an Open Researcher and Contributor ID²⁶ or ORCID prior to requesting dataset access.

Because these governance questions are specific to the requested datasets, the volume and contents of these questions will be request-specific and until the requested datasets have been entered in the dataset request form, no dataset-specific governance questions will be presented.

5. Attestations

The investigator/requesting institution attest to a set of governance-determined declarations. Attestations will affirm topics such as:

- Collaboration between the investigator and requesting institution on the study proposed in this request package
- Completion of all the requesting institution's required pre-submission activities
- Review of the governance requirements for each of the requested datasets
- The requesting institution's compliance with all applicable federal and state laws
- Assurance of a safe and secure linkage implementation that protects individual privacy and minimizes risk
- The accuracy and truthfulness of the request package

Attestations aim to affirm the collaboration between the investigator and requesting institution, verify the accuracy of information in the request package, and confirm that the investigator/requesting institution has done the expected preparatory activities.

6. Request Package Affirmation Submission

Before submission, the investigator/requesting institution affirm that the request package is complete and ready for submission. This activity occurs only when all other steps are completed and is the last step before submission. The purpose of this step is to ensure that the investigator/requesting institution approve of the request package that will be submitted.

Information Collected in the Dataset Request Form

Within a request package, a dataset request form collects information about the proposed study. Descriptions of each type of information gathered are included below.

Study Information

Study information includes the proposed study topic, aims and objectives, timing, and funding source as well as analytic methods, inclusion and exclusion criteria, anticipated limitations, use of artificial intelligence (AI) models, and plans for disseminating data and sharing findings from the study. Dataset holders will use study information to understand the rationale and intent of the research that the investigator/requesting institution aims to complete.

²⁶ Open Researcher and Contributor ID: <https://orcid.org/>

Organization and Individual Information

Organization and individual information includes organizational names and affiliated individuals, contact information, and roles on the project. Members of the study team are named and identified and a primary point of contact for the study and institution are identified. Recognizing that dataset linkage can include third-party organizations, such entities external to the requesting institution but handling data must be included. Dataset holders will use organizational and individual information to understand the entities that are involved and handling data as well as individual affiliations.

IRB and Privacy Board Information

IRB and Privacy Board information includes the identity of the IRB and/or Privacy Board who evaluated the study and their contact information. Dataset holders will use IRB and Privacy Board information in concert with study and administrative documents to confirm that the study has been appropriately reviewed for ethics and privacy and that the necessary approvals have been granted.

Requested Datasets and Linkage Implementation

For each requested dataset, the dataset request form will include general data source information, specifications for the needed population, time period, and data elements needed for the study. The dataset request form will also include proposed methods for dataset access, sharing, storage, and disposition. For the linkage implementation, the dataset request form will collect linkage method and rationale, who will do the matching, and what outputs will be produced. Dataset holders will use this information to understand what data is being requested from their dataset, what other datasets their data will be linked to, and how linkage will occur. Dataset holders will use the linkage implementation information to assess the technical feasibility of the linkage and the likelihood that the proposed linkage will be successful.

Investigators may design a study where a dataset they have generated or already have from a previous study will be linked to other requested datasets. When this is the case, one of the requested datasets in the dataset request form is their own and their contribution of the dataset to the study is noted.

Linked Dataset Creation, Access, Use, and Disposition

Linked dataset information includes who will create the linked dataset, what variables (including identifiers) will be included in the linked dataset, who will have access to the linked dataset, how the linked dataset will be transmitted to the investigator/requesting institution, and how the linked dataset will be used and disposed of after completion of the study. If the investigator/requesting institution plans to disseminate the linked dataset further in its original or modified (e.g., de-identified) form, this information would be provided in lieu of the disposal information. Dataset holders will use this information to understand the contents of the linked dataset that will be analyzed in the study, assess its sensitivity and if the proposed management plan is in alignment with the governance rules for their dataset, and evaluate if proposed disposal or dissemination is acceptable.

Privacy and Security Information

Privacy and security information collected describes how the investigator/requesting institution will preserve the privacy of the individuals whose data are in the requested datasets and linked dataset. It also describes security controls and protections that will be used throughout the data lifecycle at each organization handling the requested datasets or the linked dataset. Dataset holders will use this information to evaluate if the proposed methods align with their standard and policies regarding preserving privacy and ensuring security.

Additional Considerations

The investigator/requesting institution may consider the following points when creating and submitting a request package:

- The request package includes a large amount of varied information, making completing a request package a potentially time-intensive endeavor that likely requires input from multiple individuals from the requesting institution and inputting information over multiple sessions.
- The investigator/requesting institution should read and understand the entire request package at the outset to determine which individuals need to be involved, what additional internal review may be required prior to submission, who will complete which parts of the request package, and who will submit the request package.
- Administrative information about the study team and participating organizations should be gathered proactively. If not done systematically, collecting this type of information can take valuable time and delay submission.
- Thoroughly developing and completely documenting the methods for the proposed study and linkage may reduce the time required to complete the request package. Delays and inefficiencies may occur when the investigator/requesting institution encounters a question about the linkage or the study that they have not already considered and requires input and feedback from others. The investigator/requesting institution is encouraged to collaborate with linkage experts when designing the linkage implementation for the study and to consult with privacy and security experts to design the safest and most secure linkage options.

Appendix E. Rule Harmonization Protocol

Introduction

This appendix describes a protocol for rule harmonization, a new and structured process within the Common Governance Process to align and integrate governance rules across multiple datasets when linking individual-level health data for research. By identifying and reconciling overlapping or conflicting rules, rule harmonization supports legal, ethical, and organizational compliance; minimizes risks; and promotes responsible data use. The goal of rule harmonization is to create a single, streamlined consolidated rule set that provides clear guidance on how the linked data may be created, accessed, shared, used, and disseminated, enabling investigators and dataset holders to advance research while maintaining compliance with the policies governing the originating datasets.

Background

What Is a Governance Rule?

A governance rule is a specific directive that dictates how data is accessed, shared, linked, used, and/or disseminated²⁷ within or between organizations. Governance rules are built on three core concepts: the origin (where they originated from such as an organizational policy), the intent (the objective they aim to achieve, such as ensuring the security of individual-level data), and the application (who and what action the rule applies to such as how an investigator/requesting institution may access data).²⁸

Governance rules are established by policies and originate from a variety of sources, including laws; regulations; institutional policies; project-, study-, or dataset-specific policies; ethical requirements; and organizational practices and norms. Each source can hold one or multiple policies that assert a range of rules from broad principles for overall data management to highly specific requirements targeting a single aspect of the data lifecycle.

Because policies are established at multiple levels – federal, state, local, institutional, and project-specific – datasets and their dataset holders are subject to overlapping and interacting layers of governance rules. Some rules may align across datasets, while others may impose contradictory requirements that cannot be simultaneously satisfied.

What Is Rule Harmonization?

When datasets are linked, the health data about individuals from separate originating datasets is combined in a resulting linked dataset. The governance rules associated with originating datasets remain applicable to the linked dataset, but there is no process to combine and integrate those rules into a single, cohesive set of rules governing the linked dataset. While some governance rules across datasets may overlap, others may conflict, creating challenges for compliance, consistency, and responsible data management. Without harmonization, the investigator/requesting institution may encounter multiple, potentially contradictory rules with no

²⁷ The data lifecycle refers to the process of accessing, sharing, linking, using, and disseminating data that is fundamental to research with linked data.

²⁸ These three concepts are further explained in [Background: Foundation of Governance Rules](#).

clear framework for integration.

Rule harmonization, as proposed in the Common Governance Process, is intended to provide a structure to systematically align and reconcile governance rules across originating datasets to guide the investigator/requesting institution in responsible linked data access, sharing, linkage, use, and dissemination. From the perspective of the dataset holder, rule harmonization is a means to help dataset holders collaborate and attempt to resolve overlaps and conflicts so they may permit dataset access for linkage.

Rule harmonization seeks to model many established legal practices. For example, in corporate law, contract integration and consolidation are used to reconcile terms from multiple agreements into a single governing document. In federal systems with overlapping jurisdictions (e.g., U.S. federal and state laws), “conflict of laws” analysis is used to reconcile competing legal rules. These legal practices provide the conceptual foundation for rule harmonization.

Rule harmonization begins with the review of governance sources, policies, and rules, followed by the identification of overlaps and/or conflicts, and harmonization to – where possible – resolve overlaps and conflict. The harmonized rules are incorporated into a consolidated rule set that is reviewed, revised if needed, and approved by authorized representatives of all dataset holders involved in the linkage. Once approved by dataset holders, the consolidated rule set is then reviewed by the investigator/requesting institution, which must accept it in order to move forward to the next steps in linking datasets.

The consolidated rule set serves as the foundation for managing the resulting linked dataset, offering clear guidance to investigators/requesting institutions on how the data may be accessed, shared, linked, used, and disseminated in compliance with applicable laws, ethical standards, and policies. The consolidated rule set also serves as a basis for formal agreements that define the terms and conditions for creating and using the linked dataset, such as data use agreements or DUAs.

Why Is Rule Harmonization Necessary?

The governance rules that apply to each originating dataset that are to be linked are inherited by the linked dataset because governance rules follow the data.²⁹ To manage inherited rules that flow from multiple sources, investigators/requesting institutions need a process to gather and reconcile rules from originating datasets into a single, consolidated set of governance rules that harmonizes conflicting and overlapping rules.

Rule harmonization is important when linking multiple datasets because as the number of datasets being linked increases, so can the number of applicable governance rules and the complexity of governance. Each additional dataset’s unique set of rules increases the volume of rules to sort through and the likelihood of conflicts and overlaps.

²⁹ For example, under the Health Insurance Portability and Accountability Act (HIPAA), Protected Health Information (PHI) retains its privacy and security requirements even when combined with other datasets. If PHI is linked with non-healthcare data, the resulting dataset must still comply with HIPAA’s privacy and security rules. Similarly, a DUA for healthcare data may require that the recipient adhere to HIPAA or other privacy regulations, even if the data is combined with unrelated datasets.

Who Conducts Rule Harmonization?

The coordinating body leads the rule harmonization process with participation from the dataset holder(s)³⁰ and the investigator/requesting institution.³¹

When linking multiple datasets, neither the investigator/requesting institution nor the dataset holder(s) are well positioned to lead rule harmonization. Dataset holders are bound by their own policies, which may limit their ability to objectively reconcile rules across multiple datasets. Similarly, while the investigator/requesting institution is central in the research process, their vested interest in the outcomes of the linkage and the study could introduce bias into the harmonization process. Additionally, dataset holders and the investigator/requesting institution often face practical constraints, such as limited time and resources to identify rules from other Dataset Holders, reconcile conflicts between governance rules, and convene discussions to negotiate rules.

The coordinating body is a neutral entity. It is best suited to lead rule harmonization. Operating independently of both the dataset holder(s) and the investigator/requesting institution, the coordinating body promotes impartiality and avoids concerns of bias. Its responsibilities include facilitating collaboration, mediating conflicts, and aligning governance rules with legal, ethical, and institutional policy requirements. To fulfill these duties, the coordinating body must possess expertise in research processes, ethics, data governance, health policy and other necessary policies, and legal interpretation.³²

While the coordinating body leads the effort, rule harmonization is necessarily collaborative. Active engagement throughout this process and input from the dataset holders is essential to address concerns, refine rules, and build trust and accountability among all parties.

Common Governance Process Activities that Precede Rule Harmonization

The Common Governance Process has three stages: Intake, Screening, and Rule Harmonization. Before harmonizing governance rules, several activities establish the optimal conditions for an efficient Common Governance Process. The investigator/requesting institution must have designed the study, secured approvals from oversight entities like IRB and the Privacy Board and submitted a complete request package that consolidates study information and initiates the Common Governance Process (Intake stage, [Section 4.1](#)). The request package submission creates the record and starts the workflow within which rule harmonization will occur. The request package identifies requested datasets to be linked, and the governance

³⁰ Linking multiple datasets means involving multiple Dataset Holders in a governance process.

³¹ See [Section 5](#) description of these roles.

³² The rule harmonization process recognizes widespread interest in how artificial intelligence can streamline governance by performing key governance tasks. For a number of reasons, rule harmonization as described in this document cannot be performed by a large language model (LLM) at this time. LLMs lack the nuanced understanding required to interpret complex legal, ethical, and institutional requirements. Governance rules often involve subjective judgment, contextual considerations, and negotiation among stakeholders, elements that cannot be fully captured or resolved by an automated system. For example, reconciling conflicts between rules may require weighing competing priorities, understanding the intent behind specific policies, or addressing ethical concerns that demand human empathy and discretion. These tasks are inherently human and require collaboration, critical thinking, and stakeholder engagement.

requirements already reviewed by the investigator/requesting institution. By identifying datasets to be linked, the request package also identifies the dataset holders that will participate in rule harmonization and the sets of governance rules that will be harmonized.

Because rule harmonization is a resource-intensive process, the dataset holders conduct a preliminary "go/no-go" review to assess whether the study is feasible and permissible (Screening stage, [Section 4.2](#)) before proceeding to the Rule Harmonization stage ([below](#)). Requests that receive a "go" decision from all dataset holders during Screening advance to rule harmonization. Those that are "no-go" may receive feedback through the coordinating body but will not proceed further.

Rule Harmonization Steps

This section describes the steps for conducting rule harmonization by the coordinating body:

1. Assemble and organize a complete inventory of requested datasets' applicable governance rules;
2. Map and categorize governance rules, including identifying those that require harmonization;
3. Draft a single set of rules that reflects the governance rules from the originating datasets and resolves rule overlaps and conflicts;
4. Socialize draft consolidated rule set with dataset holders and resolve any issues of concern through collaboration and cooperation; and
5. Issue a final approval or denial of the consolidated rule set and request.

Rule harmonization business process diagrams depicting the activities within each of these steps are presented in Figure 11, Figure 12, and Figure 13³³ and sections that follow describe each step in detail. Within the Rule Harmonization stage, the dataset holders and the investigator/requesting institution make four key governance decisions (decisions 2-5 in the Common Governance Process):

- Governance Decision 2: Is the draft rule inventory complete and accurate? (dataset holder)
- Governance Decision 3: Is the draft consolidated rule set acceptable? (dataset holder)
- Governance Decision 4: Can the study comply with the consolidated rule set? (investigator/requesting institution)
- Governance Decision 5: Is the request approved? (dataset holder)

Each decision point is presented in the business process diagrams with a yellow diamond and described in detail below. White diamonds reflect operational decisions that support navigation within the rule harmonization process and are represented distinct from governance decisions because they lack governance implications for the request.

³³ Explanations for the Intake and Screening stages, as well as their business process diagrams, are found in [Section 4](#).

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Step 1. Identify and Review Governance Sources

The first step in rule harmonization (see Figure 11) is for the coordinating body to gather from the dataset holders the relevant governance sources that apply to the requested datasets (e.g., policy documents, informed consents, contracts, and agreements such as DUAs).³⁴ Relevant sources contain, for example, policies about research, data use, access, and/or linkage. The coordinating body may have to review a source's content to determine relevance and discard irrelevant sources. Recognizing that governance sources can vary widely in how they are created, structured, and published, a dataset may have just a few or many governance sources, each containing one or more policies and rules. Because governance sources often reference other sources, such as a DUA requiring users of a dataset adhere to HIPAA, the coordinating body also gathers and reviews referenced governance sources. This step is repeated for each requested dataset.

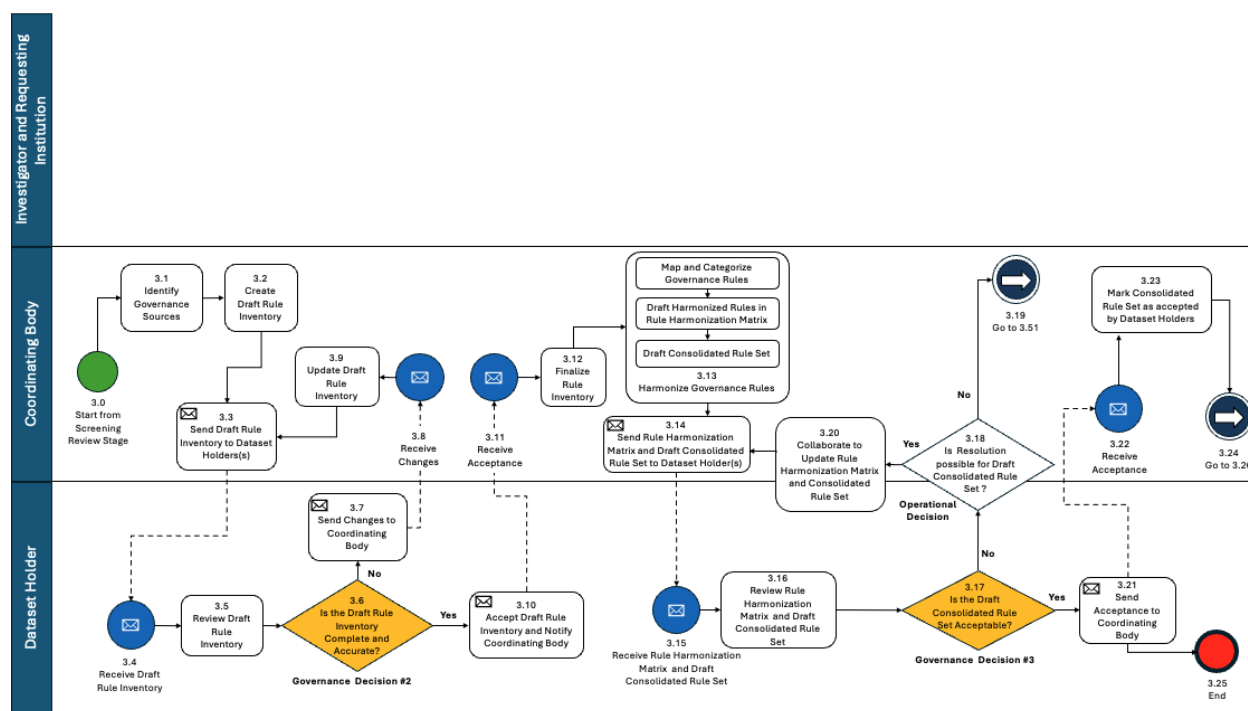


Figure 11: Rule Harmonization Business Process Diagram (1 of 3)

The coordinating body catalogs each governance source by recording information about its identity and provenance (e.g., name, version number, date, and information about where it was located). As governance sources are updated over time, recording provenance information provides important context on the state of governance at the time of rule harmonization.

Step 2. Inventory and Validate Governance Rules

Once governance sources have been collected, the coordinating body carefully reviews each source's content and inventories the rules it contains. This involves extracting and documenting

³⁴ Some governance sources may not be applicable to research linkage; for example, policies on disaster recovery or system backups are unlikely to be relevant to an Investigator requesting access to a dataset.

the research- and linkage-relevant rules³⁵ from each governance source. The coordinating body enters rules into a rule inventory in a standard format using a structured worksheet that retains each rule's originating source (see [Template 1](#)) and assigns each rule to a domain and subdomain. This protocol recommends using each action in the data lifecycle as domains, and relevant subtopics within that action as subdomains; for example, data access as a domain and access controls as one subdomain of data access (see [Template 2](#)). Every domain and subdomain may have zero, one, two, or many rules and these patterns will vary by dataset. By organizing rules the same way across datasets, like rules are grouped and rules can be easily compared across datasets in the next step. One possibility is the identification of internal rule conflicts, where multiple rules for the same dataset from different governance sources are in conflict with each other.

The coordinating body completes this step and creates a rule inventory including rules for all requested datasets. The rule inventory is considered a draft until accepted by the dataset holders: the coordinating body shares the draft rule inventory with each dataset holder to confirm that the governance sources are current, the inventory is complete, and any interpretation of rules (when necessary) is accurate. The process is now at Governance Decision #2: Is the draft rule inventory complete and accurate? Each dataset holder either accepts or does not accept the draft rule inventory as accurate and complete. When not accepted, the dataset holder modifies the rule inventory for completeness and accuracy. Once the rule inventory is accepted by the dataset holders, the coordinating body updates the rule inventory as final with the understanding that dataset holders may provide further feedback and changes as needed.

Step 3. Map and Categorize Rules for Harmonization

Once the rule inventory is finalized, the coordinating body compares rules across the requested datasets by domain and subdomain, mapping rules across requested datasets that must work together using a rule harmonization matrix (see [Template 3](#)). Because a dataset may have zero, one, or multiple rules in a domain, the mapping exercise may be a comparison of two, three, or more rules. The coordinating body categorizes the groups of governance rules based on relationships between rules and identifies areas that require harmonization as follows:

- **Duplicate Rules:** Rules that are identical across datasets. These rules are deduplicated and one flows through unchanged to the consolidated rule set.
- **Unique Rules:** Rules that apply to only one dataset and do not align with rules from other datasets. These rules flow through unchanged to the consolidated rule set.
- **Overlapping Rules:** Rules that are semantically different but materially the same: they may use different wording, phrasing, or other specifics but impose comparable requirements or achieve the same outcomes. For example, one dataset may require that data be stored in "secure, access-controlled environments," while another specifies that data must be stored in "restricted systems with role-based access controls." Although phrased differently, both rules describe the same underlying access/security requirement. Overlapping rules require harmonization to ensure clarity for investigator and/or requesting institution obligations while eliminating – where possible – duplicative requirements.

³⁵ This means that not every governance rule from every governance source will be included in the inventory

- **Conflicting Rules:** Rules that impose contradictory requirements that cannot be simultaneously satisfied. For example, one dataset may require data to be encrypted using a specific algorithm, while another may require a different encryption standard for compliance with its governance policies. Conflicting rules require harmonization to find a middle ground or they may be determined to be impossible to link.

The mapping and categorization result in a matrix of rules organized by domains where each row of similar rules is assigned to one or multiple categories based on how they related to one another. The assigned categorization directs the coordinating body on what to do next. Duplicate and unique rules will simply flow into the consolidated rule set. Overlapping and conflicting rules will be harmonized in the next step.

Detailed documentation of rule mapping and categorization is critical for maintaining transparency and provenance throughout the harmonization process. Dataset holders may reference mapping documentation when evaluating how their rules were categorized and harmonized with others. This documentation will serve as a valuable reference for replicability, updates, or modifications to the governance framework, ensuring that future harmonization efforts can build on prior decisions and insights.

Step 4. Draft Harmonized Rules and Consolidated Rule Set

For overlapping and conflicting rules, the coordinating body drafts harmonized rules for consideration by the dataset holders:

- For overlapping rules, the coordinating body may select one of the overlapping rules that may be acceptable to all dataset holders or draft a new harmonized rule that serves as a middle ground, blending aspects of the rules that overlap.
- For conflicting rules, the coordinating body drafts a single harmonized rule that may reflect the shared intent of the conflicting rules and aims to be acceptable by dataset holders. Conflicting rules may require the Coordinating Body to consult the policies and governance sources to consider how much flexibility is available.

Harmonized rules should balance the needs of all dataset holders and be grounded in the framework's guiding principles and standards. When possible, the coordinating body will reference rule harmonization materials from past successful linkages, where similar rule conflicts were overcome. The coordinating body then drafts a consolidated rule set by integrating harmonized, duplicate, and unique rules organized by actions in the data lifecycle (e.g., rules about data access).

Step 5. Engage Dataset Holders about Acceptance of the Consolidated Rule Set

The coordinating body shares with the dataset holders the draft rule harmonization matrix ([Template 3](#)) and draft consolidated rule set. Sharing all materials allows the dataset holders to review the rule categorization and engage with the harmonized rules, collaborating and cooperating to – as much as possible – reach agreement on harmonized rules that will allow data linkage to occur. The process is now at Governance Decision #3: Is the draft consolidated rule set acceptable? The precise criteria for acceptability will vary between Dataset Holders but generally means containing rules that are clear, enforceable, compliant with applicable policies, and ready for implementation. Each dataset holder reviews the draft consolidated rule set and

either accepts or does not accept the draft consolidated rule set. As part of review, other individuals within the dataset holder's organization such as technical experts may evaluate compatibility with the systems and processes used to manage the linked dataset, while legal counsel may be needed to evaluate the harmonized rules for compliance with applicable laws. Once the dataset holders agree that the rules are clear, enforceable, and ready for implementation, they accept the draft consolidated rule set. By accepting, the dataset holder is also affirming their willingness to sign a DUA with these rules included as terms. Once the coordinating body has received acceptances from all dataset holders, the consolidated rule set is no longer a draft, but it is not yet finalized.

In cases where harmonized rules are not acceptable, the coordinating body should facilitate exchanges between dataset holders to seek solutions, modifying harmonized rules based on feedback from each dataset holder until the consolidated rule set is accepted by all dataset holders, or a determination is made that it is not feasible to achieve a consolidated rule set. Discussions between dataset holders are facilitated by the coordinating body, which is responsible for facilitating this collaborative and iterative process, ensuring that discrepancies or conflicts are resolved. Dataset holder cooperation and collaboration may occur in asynchronous communication or as a group.

In cases where the consolidated rule set cannot be accepted despite efforts between the dataset holders, the coordinating body will prompt the dataset holders to deny the request.

Step 6. Confirm Study Compliance with Consolidated Rule Set

For consolidated rule sets that have been accepted by all dataset holders, the Coordinating Body sends the consolidated rule set to the investigator/requesting institution to confirm their ability to comply (see Figure 12). The process is now at Governance Decision #4: Can the study comply with the consolidated rule set?

If the investigator/requesting institution can comply, they send an acceptance to the coordinating body. During this time, the coordinating body is responsible for responding to any questions with clarifications.

If compliance is not possible, the investigator/requesting institution identifies which rules they cannot comply with and determines whether or not they will alter the study (e.g., remove a dataset) or seek a change in the consolidated rule set or if they wish to cancel the request (an operational decision in the process). The investigator/requesting institution then sends the feedback and decision to the coordinating body. Determining whether or not to cancel may depend on the type of rule conflicts in the consolidated rule set.

When rule feedback is received without a decision to cancel the request, the coordinating body revisits the consolidated rule set with revisions. When rule feedback is received with a decision to cancel, the coordinating body updates the request status to canceled and notifies the dataset holders.

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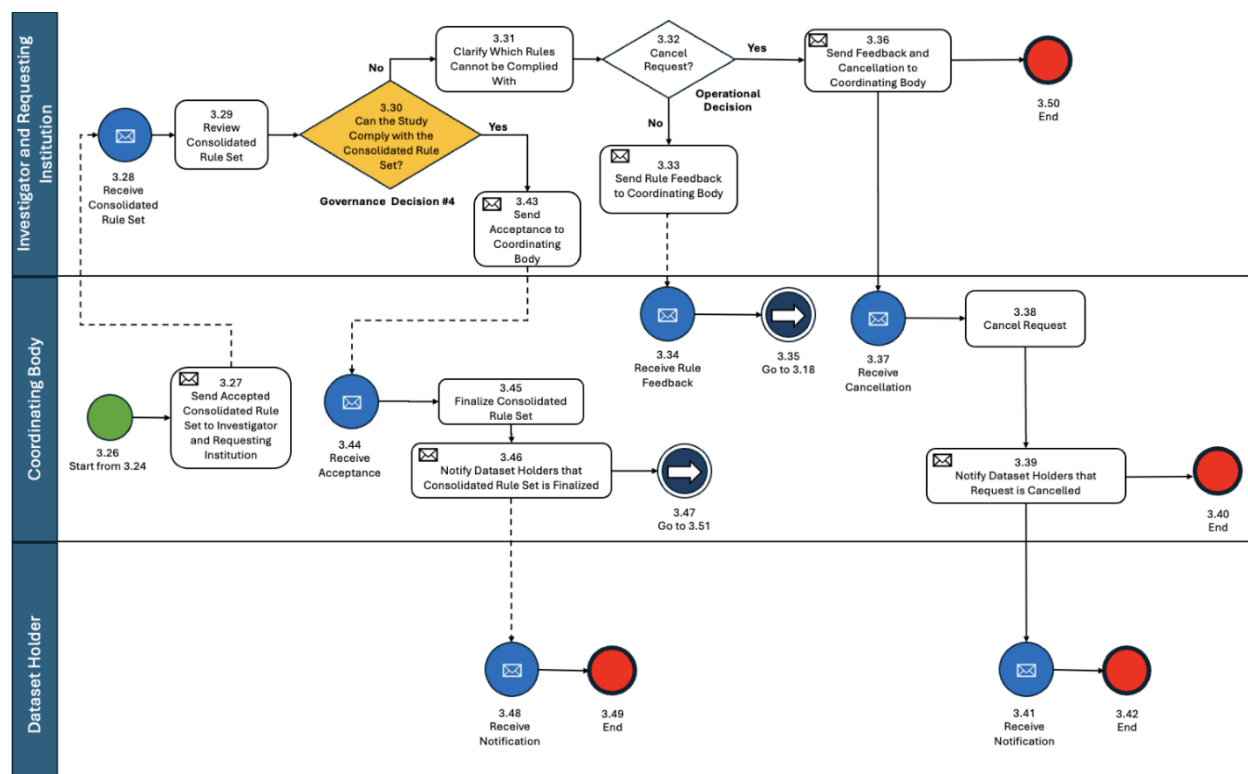


Figure 12: Rule Harmonization Business Process Diagram (2 of 3)

Step 7. Decide on the Request for Data Linkage

The dataset holders make the final governance decision to approve or deny the request to link their dataset(s) during the last step in rule harmonization (see Figure 13). The coordinating body updates the request status to ready for decision and that action notifies the dataset holders, prompting them to conduct their final request review. The process is now at Governance Decision #5: Is the request approved? This decision is the formal approval by the dataset holders to permit access to and use of their dataset for the proposed linkage under the terms set forth in the consolidated rule set.

Approve the Request: The dataset holders conduct their final review of the request package and consolidated rule set, before approving the request to proceed. For requests that have been approved in screening and have an accepted consolidated rule set, approval by dataset holders is expected.

The dataset holders send a formal approval to the coordinating body. Once approved by all dataset holders, the coordinating body marks the request as approved and notifies the investigator/requesting institution. This marks the end of the rule harmonization process for an approved request. The dataset holders may proceed with creating a DUA(s) based on the rules established in the consolidated rule set to start the study.

Deny the Request: If the consolidated rule set could not be accepted by all dataset holders and

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a resolution to eliminate rule conflicts cannot be reached, the dataset holders deny the request and share the rationale for denial with the coordinating body as request feedback. Upon receipt of a denial, the coordinating body marks the request as denied and communicates that decision and feedback to the investigator/requesting institution.

The investigator/requesting institution receive and review feedback and then determine if they can and will revise the study or the linkage implementation in order to address the rule conflict and resubmit the request package. If revision and resubmission is pursued, the investigator/requesting institution return to the Intake stage. This marks the end of the rule harmonization process for a denied request.

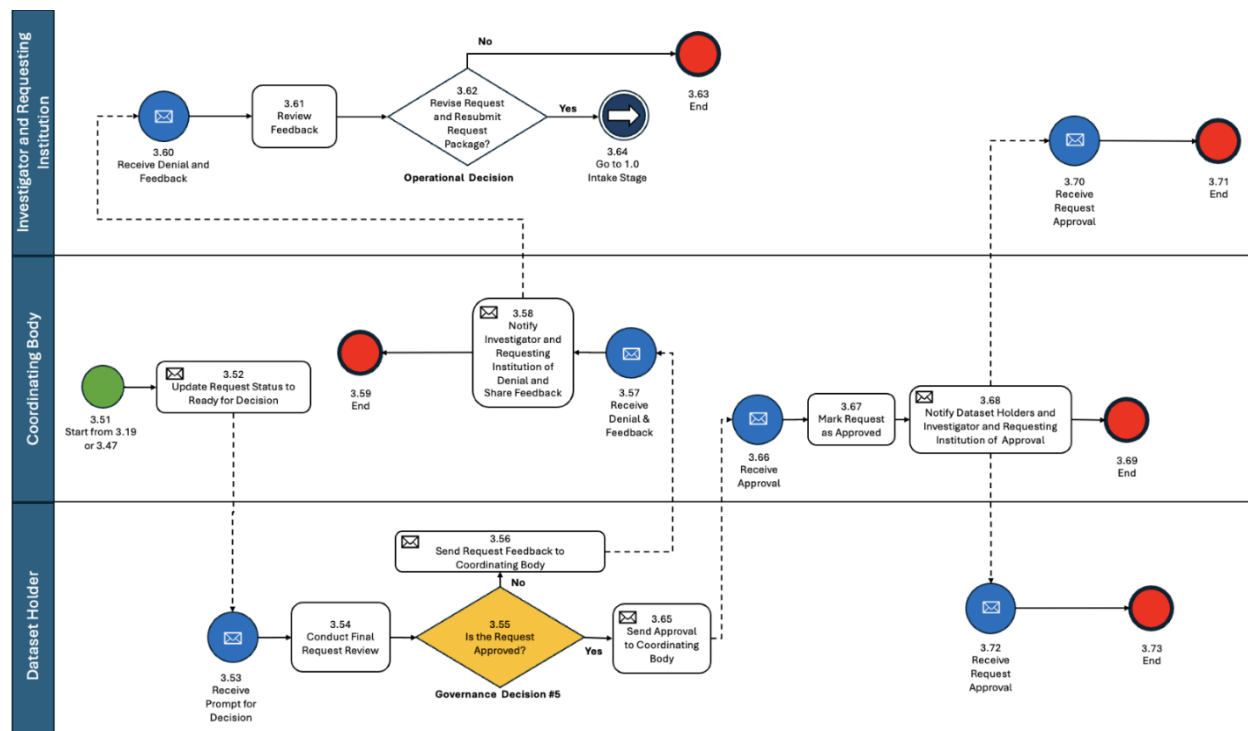


Figure 13: Rule Harmonization Business Process Diagram (3 of 3)

A denied request should be treated as an important learning opportunity for the Linkage Governance Framework. These insights can help inform future governance processes and improve the feasibility of linking datasets for research. When this occurs, the coordinating body should capture a description of why data linkage could not be accomplished and, where possible, disseminate recommendations to encourage change in dataset governance that could facilitate future linkage.

Background: Foundation of Governance Rules

Governance rules are grounded in their origin, intent, and/or application. The origin refers to the law, policy, regulation, etc. where the rule comes from. The intent refers to the purpose of the rule, and the application refers to who and how the rule accomplishes its desired intent.

Origin of Governance Rule

Governance rules can originate from a variety of types of sources including but not limited to federal, state, local, or other laws, regulations, or policies; institutional policies; project, study, or dataset policies; ethical requirements; and operational processes or norms.

Laws: Laws define legal requirements that are among the most critical considerations in rule harmonization, as they impose a variety of obligations at many different levels (e.g., obligation for an institution, a person, or an information system). Laws such as HIPAA govern the handling of PHI and may dictate strict requirements surrounding de-identification, access controls, or security measures that must be met to maintain compliance. While laws define what the requirements are, they may not necessarily dictate how the requirements are to be met. Laws also specify when and to what the law does and does not apply and exceptions to the law. In this example, HIPAA is the origin of many policies and is organized into sections that create hundreds of rules applicable to different parts of the data lifecycle. For example, a hospital's data may need to adhere to HIPAA's Privacy Rule and Security Rule, which mandate safeguards to protect sensitive information.

Laws may overlap with other laws, where one law references another or one builds on the obligations set by another. An example is state-specific laws, such as the California Consumer Privacy Act, that imposes additional restrictions beyond HIPAA, such as stricter consent requirements or limitations on data sharing.

Legal requirements are non-negotiable and must be strictly adhered to during rule harmonization. Failure to comply with legal requirements can result in severe penalties, including fines, sanctions, and reputational damage.

Institutional Policies: Institutional policies are the policies set by an organization that reflect its priorities, risk tolerance, and operational practices. These policies may include procedural requirements, such as requiring a data governance council to review and approve all external data sharing activities. For example, one institution may require data sharing to occur via its own secure file transport protocol and define the conditions that permit remote access. Organizations may have one comprehensive policy document or many more targeted policy documents. Institutional policies often build on or reference laws such as HIPAA or standards such as the National Institute of Standards and Technology (NIST) Special Publication 800-53 that defines security controls for federal information systems. For example, while all hospitals may have policies requiring compliance with HIPAA, a hospital may specify which de-identification methods permitted under HIPAA are acceptable for its datasets.

Project-, Study-, or Dataset-Specific Policies: Projects, studies, or datasets may establish unique policies that govern the data associated with a study, project, or dataset. These policies may build on institutional policies at the organization where the project study or dataset resides and may address a variety of issues specific to the project, study, or dataset such as a study's consent requirements, data usage restrictions on specific data assets, or the criteria that require additional engagement with the IRB. For example, a dataset's use may be restricted to specific topics or domains of research, prohibit use by for-profit companies, or require a specific attribution in all resulting publications.

IRB or Privacy Board Directives or Policies: Certain governance rules originate from an IRB or Privacy Board's directives or policies, informed consent documents, or processes. For an IRB, an example would be the process for notifying the IRB about an unanticipated deviation

from the study protocol. Regarding informed consents, datasets collected as part of a research study will have a dedicated consent document that stipulates rules about how the research data can be used, whereas data from routine healthcare delivery usually falls under a broader institutional consent that covers all data practices. Example consent rules from research-created data are that a person's data cannot be linked without reconsenting them for linkage on a study-by-study basis.

Operational Practices or Norms: Operational practices or norms are the informal, procedural, or culturally ingrained methods organizations use to manage data. These practices are not always formally documented but influence how governance rules are implemented throughout the data lifecycle. For example, one dataset holder may back up all their data nightly. When datasets in their environment need to be destroyed, backup copies of the dataset may also need to be considered as part of the destruction process.

Intent of Governance Rule

Every governance rule is designed to achieve one or multiple objectives, which reflect the underlying priorities of the originating policies. Intentions for governance rules may include transparency, data minimization, purpose specification and use limitation, privacy, security, consistency and standardization, accountability, and replicability.

Transparency: Transparency means that the organization is open and clear about how the data is accessed, shared, linked, used, or disseminated. This includes providing clear information about consent, data provenance, and usage restrictions. Transparency empowers data users by informing them of the purposes of data collection and their rights regarding its use. It also supports compliance with legal and ethical standards that require explicit disclosure of data practices.

Transparency also promotes findability and accessibility. For example, metadata that documents consent agreements, data origins, and usage restrictions makes the data easier to locate and understand, while ensuring that stakeholders are informed about its governance.

Data Minimization: Data minimization is a key directive in HIPAA³⁶ and focuses on collecting and using only the data necessary to achieve the intended purpose of the linkage. By limiting the scope of the data, this reduces the risk of privacy breaches, minimizes the potential for misuse, and ensures compliance with regulations that require proportionality in data collection and use. Data minimization also supports efficient data management, as smaller datasets are easier to secure and analyze. Data minimization complements interoperability, as smaller, well-defined datasets are easier to integrate and analyze across systems while maintaining ethical and legal constraints.

Purpose Specification and Use Limitation: Purpose specification and use limitation means clearly defining the reasons for collecting data and restricting its use to those stated purposes unless additional consent is obtained. This principle prevents unauthorized or unethical uses of data, aligns linkage with original objectives, and complies with applicable regulations. By adhering to defined purposes, organizations safeguard the integrity of the data and reduce risks

³⁶ HIPAA Minimum Necessary Requirement: <https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/minimum-necessary-requirement/index.html>

associated with misuse.

Privacy: Privacy refers to the right of individuals to control how their personal health data is collected, used, shared, and stored. It ensures that sensitive health details, such as diagnoses, treatments, and medical history, are protected from unauthorized access or disclosure. Health data privacy involves safeguarding confidentiality while respecting patients' autonomy and their ability to make informed decisions about who can access their information. Privacy is also about ensuring individuals feel safe sharing details about their medical history and conditions without fear of judgment or misuse. Safeguarding health data privacy helps prevent unauthorized access, reducing the risk of identity theft, discrimination, or exploitation based on medical information. Examples of privacy rules include data anonymization and small cell suppression. For example, privacy-enhancing techniques, such as de-identification or encryption, must be applied consistently across the linked dataset to maintain compliance with applicable regulations and ethical standards.

Security: Security involves implementing measures, policies, and technologies designed to protect data from unauthorized access, breaches, loss, or misuse. It safeguards the confidentiality, integrity, and availability of data throughout its lifecycle, using security measures such as encryption, access controls, and regular audits. Security also enhances accessibility, by implementing mechanisms so that only authorized users can access data safely while preventing unauthorized access or breaches. Security rules focus on safeguarding data from unauthorized access, breaches, or misuse. These rules may include requirements for encryption, access controls, regular audits, or secure storage environments. For example, one dataset may require data to be stored in encrypted formats, while another may mandate additional physical security measures for servers.

Consistency and Standardization: Consistency and standardization refer to the alignment of governance rules and technical requirements to ensure seamless data integration, sharing, and management across datasets. This involves harmonizing elements such as data formats, encryption protocols, and interoperability standards to create a unified framework that minimizes errors and inconsistencies. For example, one dataset may store information in a particular file format, while another uses a widely accepted open standard.

Accountability: Accountability refers to the obligation of investigators, organizations, and systems to take responsibility for their actions, decisions, and compliance with established rules and standards. It involves defining roles, responsibilities, and enforcement procedures along with creating mechanisms to monitor, enforce, and report adherence to governance frameworks. By holding all parties answerable for their roles in managing and protecting data, accountability fosters a culture of transparency, ethical behavior, and responsibility in data-sharing initiatives. Accountability also supports findability, as clear documentation of roles and responsibilities helps stakeholders identify who is responsible for specific aspects of data governance and management.

Replicability: Replicability refers to the ability to consistently replicate a research study or governance process and reproduce research findings by ensuring that data management practices, processes, and documentation are standardized, transparent, and well-maintained. It involves establishing clear protocols for data collection, storage, sharing, and analysis so that other researchers can access and use the same data in a reliable and repeatable manner. Rules about replicability ensure that governance processes are open and well-documented,

allowing others to understand how data was handled and analyzed. It supports verification of the integrity of research findings by enabling others to reproduce results using the same data and governance practices. Replicable governance supports seamless sharing and re-use of data across research teams, fostering collaboration and accelerating scientific progress.

Application of Rules: Governance rules can apply to different parties (persons or organizations) and to different actions that party may take. In linking data for research, governance rules can apply to one or more steps in the data lifecycle: access, sharing, linkage, use, and dissemination.

Data Access: Data access is acquiring data from a data repository or other system to store, manage, and share data for reuse. Authorized persons or entities may gain entry to datasets under governance rules that outline eligibility criteria, approval processes, and monitoring requirements. These rules define who can access the data, under what conditions, and through what mechanisms, ensuring compliance with security and governance requirements while protecting sensitive information. For example, one dataset may require access to occur only within secure environments, while another may permit remote access with multi-factor authentication or transmission of data to an investigator on physical media.

Data Sharing: Data sharing refers to making data available to the broader data user community, such as submitting data to a repository for dissemination or sending data to an investigator for an approved research study. It involves granting access to data between entities under governance rules that specify recipients, conditions, mechanisms, and restrictions. These rules define who data can be shared with, the conditions for sharing, the parameters for safe and secure transmission, and the safeguards required to ensure compliance with privacy, ethical, and legal standards. For example, one dataset may limit sharing to academic institutions, while another may permit sharing with government agencies under specific agreements.

Data Linkage: Data linkage refers to the process of combining data from two or more datasets at the individual level to create a unified dataset. Rules governing data linkage may define if data can be linked, processes for requesting linkage, what linkage methods are permissible, required technical standards, and what parameters are required when conducting linkage. For example, a hospital may allow linkage only using Privacy-Preserving Record Linkage so that no identifiers are shared and prohibit re-identification of the individuals within the linked dataset.

Data Use: Data use refers to working with data for secondary research or other analytical purposes. Rules on data use can specify how linked data can be used, permissible research purposes, and restrictions on analysis. For instance, one dataset may limit use to disease-specific research, while another may allow broader applications. Since data use reflects a sequence of tasks including data management, storage, and retention, data use rules can dictate how long data can be kept and under what conditions it must be deleted. For example, one dataset may require retention for the period of the study, while another mandates deletion after a specific timeframe.

Data Dissemination: Data dissemination refers to the process of distributing data used in or created from research in its original or derivative form to the public, including other investigators, clinicians, and other partners, and is distinct from dissemination of research findings. In the case of linkage, dissemination refers to distribution of the originating datasets and linked dataset.

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Rules govern how data can be additionally distributed to authorized parties or the public, including the conditions, methods, and safeguards required to ensure compliance with privacy, ethical, and legal standards. For example, some linked datasets will be prohibited from further distribution while others will be created with extensive reuse as an explicit goal such that intentional rules and processes will dictate how the linked dataset can be disseminated for widespread use.

Template 1. Rule Inventory

The rule inventory provides a format to create a detailed record of governance rules, organized by domain and subdomain. This template (see Table 5) ensures that all rules are documented systematically, with clear references to their governance sources and provenance.

Table 5: Rule Inventory Template

Rule Domain	Rule Subdomain	Dataset	Rule	Governance Source	Source Link	Source Details	Notes
Data Access	Access Controls	[insert dataset name]	Data must be stored in secure environments.	Institutional Policy	Link; 06-15-2025	Policy Version 2.0, Section 4.3, Updated 2024	Aligns with HIPAA Security Rule
Data Sharing	Recipients	[insert dataset name]	Data may be shared only with government agencies.	DUA	Link; 06-15-2025	DUA Version 1.1, Section 2.1, Signed 2024	Stricter sharing rules than Dataset B
Data Use	Permissible Uses	[insert dataset name]	Data may be used only for public health research.	Informed Consent Form	Link; 06-15-2025	Consent Form ID #12345, Approved by IRB 2023	Stricter sharing rules than Dataset B

Complete the columns in the rule inventory template as follows:

- **Rule Domain:** The broad governance area the rule falls under (e.g., Data Access, Data Sharing, Data Linkage, Data Use, Data Dissemination).
- **Rule Subdomain:** A more specific topic within the domain (e.g., Access Controls, Recipients, Permissible Purposes).
- **Dataset:** The dataset to which the rule applies.
- **Rule:** The specific directive or guideline governing the dataset (e.g., "Data must be stored in secure environments"). The rule should be written verbatim or paraphrased from the governance source for accuracy.
- **Governance Source:** The source from which the rule originates (e.g., federal law, institutional policy, DUA, consent form).
- **Source Link:** Hyperlink for where the source was retrieved from and accessed date.

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- **Source Details:** Additional information about the governance source, such as version/date, relevant sections, and any references to external policies.
- **Notes:** Contextual information or observations about the rule (e.g., "Aligns with HIPAA Security Rule," "Stricter sharing than Dataset A").

Template 2. Proposed Rule Harmonization Domains and Subdomains

The rule harmonization domains and subdomains provide a proposed method for organizing governance rules across datasets. Table 6 organizes the rules into key areas of the data lifecycle: data access, sharing, use, linkage, dissemination, retention, and technical standards. Each domain is further refined into suggested subdomains to group rules by specific topics (e.g., access controls, permissible research purposes). These subdomains are not exhaustive and serve as a starting point for structuring governance rules. During the rule harmonization process, additional subdomains may emerge based on the unique characteristics of the datasets, governance sources, and specific requirements of the study.

Table 6: Proposed Rule Harmonization Domains and Subdomains

Domain	Subdomains
Data Access	Access Controls User Eligibility Access Mechanisms Access Approval Process Access Monitoring and Auditing Access Duration
Data Sharing	Recipients Conditions Mechanisms Restrictions Accountability/Oversight
Data Use	Permissible Uses Prohibited Uses Analysis Restrictions Dissemination of Results Ethical Oversight Data Use Transparency
Data Linkage	Methods Approval Feasibility Safeguards Accountability/Oversight Legal and Ethical Considerations
Data Dissemination	Recipients Conditions Mechanisms Restrictions

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	Accountability/Oversight Dissemination Transparency
Data Retention	Duration Conditions Safeguards Accountability
Technical Standards	Data Formats Encryption Protocols Interoperability Requirements System Security Data Quality Data Validation

Template 3. Rule Harmonization Matrix

The rule harmonization matrix is a tool to document the process of mapping governance rules across datasets, categorization of relationships between rules, and proposed harmonized rules. The rule harmonization matrix captures rules from originating datasets, the category that best describes how rules are related and rationale, and the proposed harmonized rule and rationale. Table 7 illustrates how the rule harmonization matrix can be completed with four examples.

Table 7: Rule Harmonization Matrix

Rule Domain – Subdomain	Dataset A Rule(s)	Dataset B Rule(s)	Data Set C Rule(s)	Mapping Category	Mapping Categorization Rationale	Proposed Harmonized Rule	Rationale for Harmonization
Data Access – Access Controls	Data must be stored in secure environments with restricted access.	Data must be stored in restricted systems with role-based access controls.	Data must be stored in secure environments with multi-factor authentication.	Overlap	All rules require secure environments but differ in specifics (e.g., restricted access, role-based permissions, authentication).	Data must be stored in secure, access-controlled environments that meet restricted access and authentication requirements.	Aligns with stricter security requirements while accommodating all datasets' primary intent.
Data Sharing – Recipients	Academic institutions only.	Academic institutions only.	Academic institutions only.	Duplicate	All datasets specify identical rules allowing sharing only with academic institutions.	Allow sharing with academic institutions only.	No harmonization needed; all rules are identical.
Data Use – Permitted Use	Public health research only.	Broad use permitted.	Public health research only.	Conflict	Datasets A and C restrict use to public health research. Dataset B allows broader use.	Restrict use to public health research.	Aligns with consent form restrictions from Datasets A and C.
Data Use – Retention	Data must be retained for a minimum of 5 years.	No retention requirements specified.	No retention requirements specified.	Unique	Dataset A has a specific retention rule, while Datasets B and C have no requirements.	Retain data for a minimum of 5 years.	Dataset A's rule flows into the Consolidated Rule Set unchanged.

Complete the columns in the rule harmonization matrix as follows:

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- **Rule Domain:** The broad governance area the rule falls under (e.g., Data Access, Data Sharing, Data Use).
- **Rule Subdomain:** A more specific topic within the domain (e.g., Access Controls, Recipients, Permissible Purposes).
- **Dataset A Rule(s):** The specific governance rule applicable to Dataset A, as documented in the rule inventory (e.g., Data must be stored in secure environments, or data must be stored in restricted systems with role-based access controls).
- **Dataset B Rule(s):** The specific governance rule applicable to Dataset B, as documented in the rule inventory.
- **Dataset C Rule(s):** The specific governance rule applicable to Dataset C, as documented in the rule inventory.
- Mapping Category:
 - **Duplicate Rules:** Rules that are identical across datasets. These rules are deduplicated and flow directly into the consolidated rule set without harmonization.
 - **Unique Rules:** Rules that apply to only one dataset and do not overlap with rules from other datasets. These rules flow directly into the consolidated rule set without harmonization.
 - **Overlapping Rules:** Rules that are semantically different but materially the same; they may use different wording, phrasing, or other specifics but impose comparable requirements or achieve the same outcomes.
 - **Conflicting Rules:** Rules that impose contradictory requirements that cannot be simultaneously satisfied.
- **Mapping Categorization Rationale:** A detailed explanation of the relationship between the rules from all datasets, noting whether they overlap or conflict. The rationale should clearly identify the specific differences or similarities between rules.
- **Proposed Harmonized Rule:** The rule proposed to reconcile the overlap or conflict among all the datasets' rules.
- **Rationale for Harmonization:** The justification for the proposed harmonized rule, explaining why it was chosen and how it aligns with governance principles, legal requirements, and ethical standards.